

PHY 517 / AST 443:

Observational Techniques in Astronomy

Lecture 4:

(Photometry)

Spectroscopy

Last Time: CCDs

You want to take a sky image with our telescope + CCD camera. Which calibration images do need to take in order to analyze the science image? How do you take them? What do you do with them?

- **Dark frames:** same exposure time, same temperate -> measures bias + dark current
- **Flat-fields:** images of a uniform surface (e.g. dome), taken with the camera attached to the telescope in the same set-up as for the science image (*also needs to be dark-corrected*)

$$\text{calibrated image} = \frac{\text{science image} - \text{dark}}{\text{flatfield}}$$

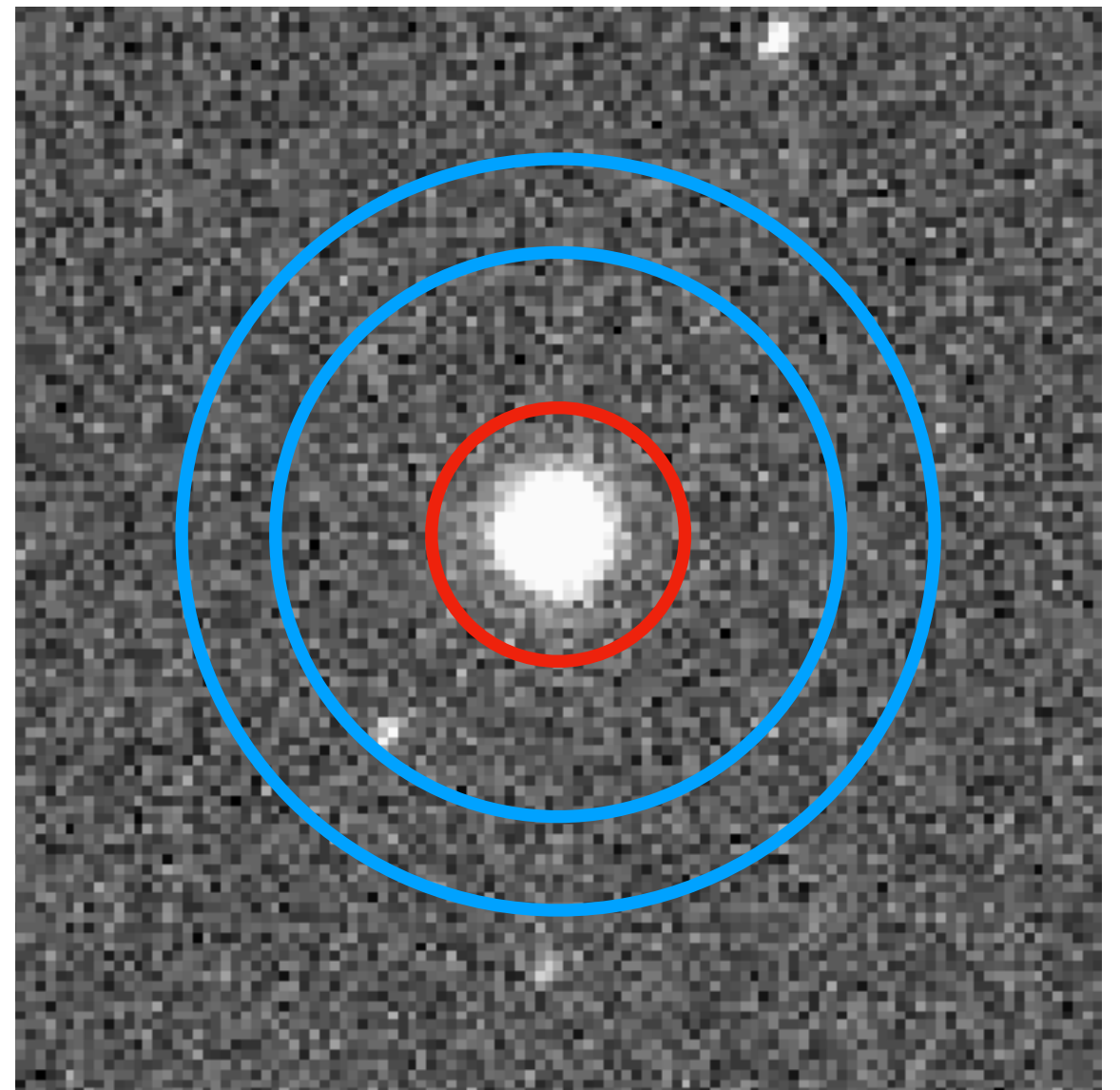
Calibrated image

Photometry: measure the brightness (also colors) of sky objects

How would you measure the flux of this star?

1. Sum up counts within aperture
2. Measure sky background (per pixel) from empty region of sky
3. Subtract estimated sky counts
4. Repeat for a reference star of known magnitude

$$m_1 - m_2 = -2.5 \log \left(\frac{F_1}{F_2} \right)$$



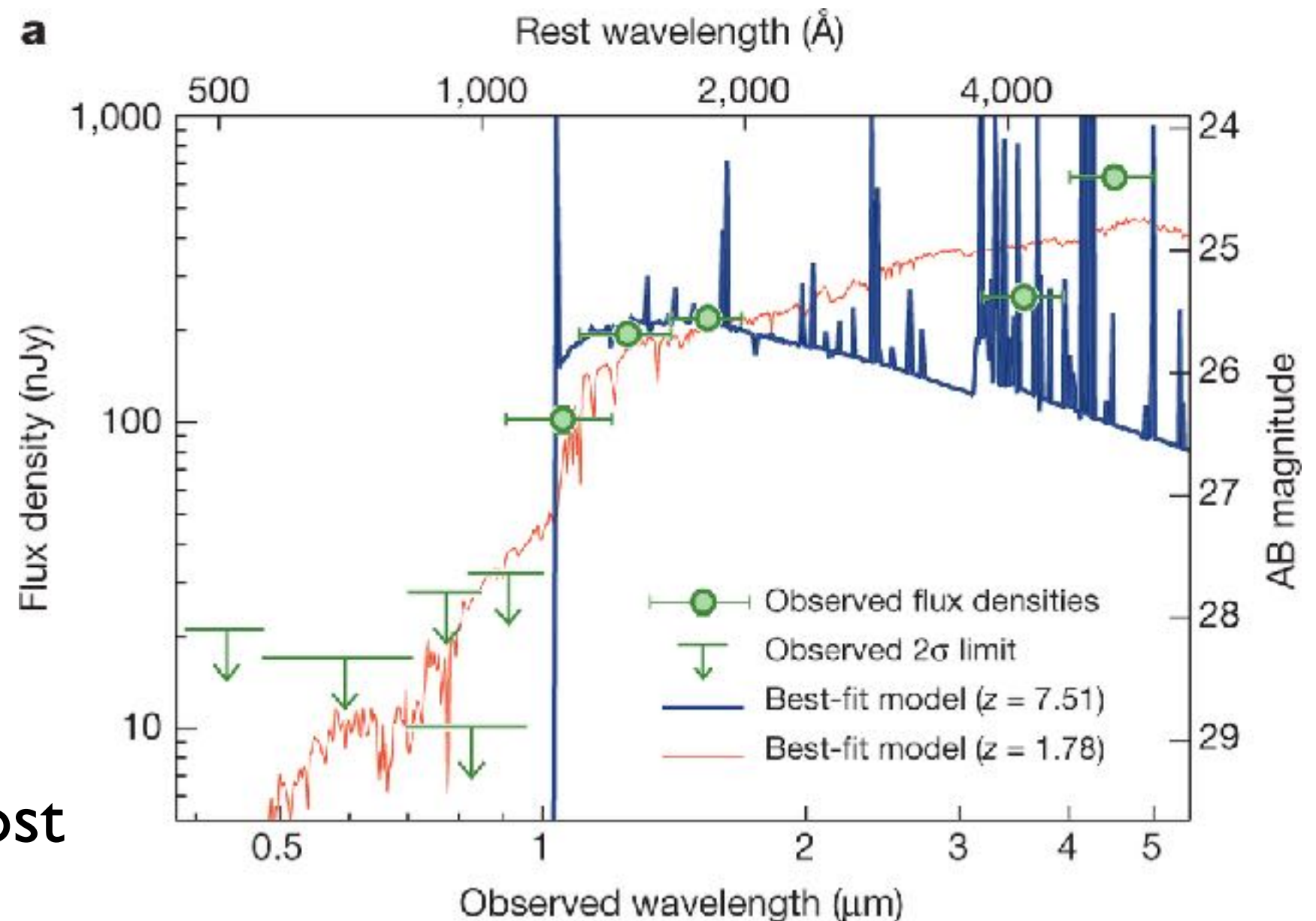
Spectroscopy

Motivation

photometry only measures integrated flux (over filter band)

gives some
information
about the object
properties, but
often not
enough

e.g.: finding the most
distant galaxies

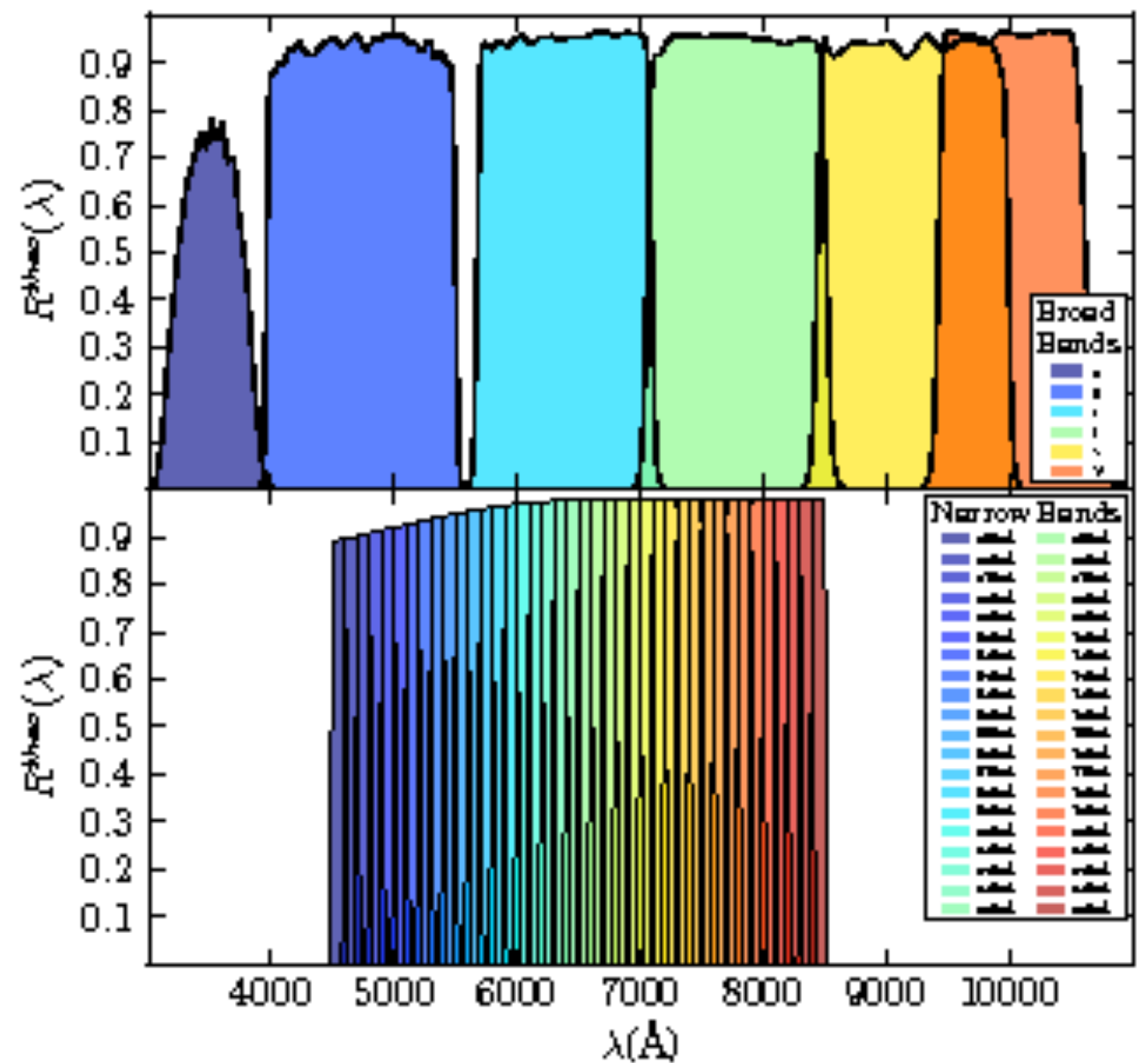


Narrow-band imaging

can determine spectrum
of object with images in
many narrow-band filters

advantage: can determine
spectra of all objects in
the same FOV

disadvantage: have to take
a lot of images!

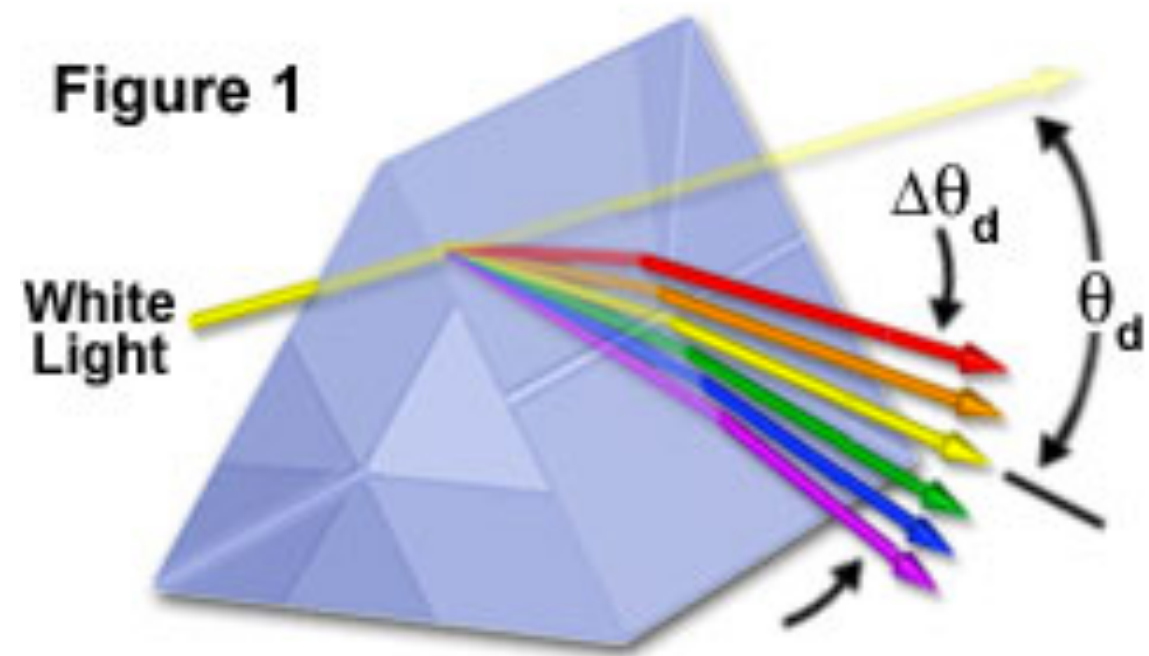


The PAU Survey

Spectroscopy

add a dispersing element
to split up the light from
an object: measure the
spectrum directly

e.g. a prism:

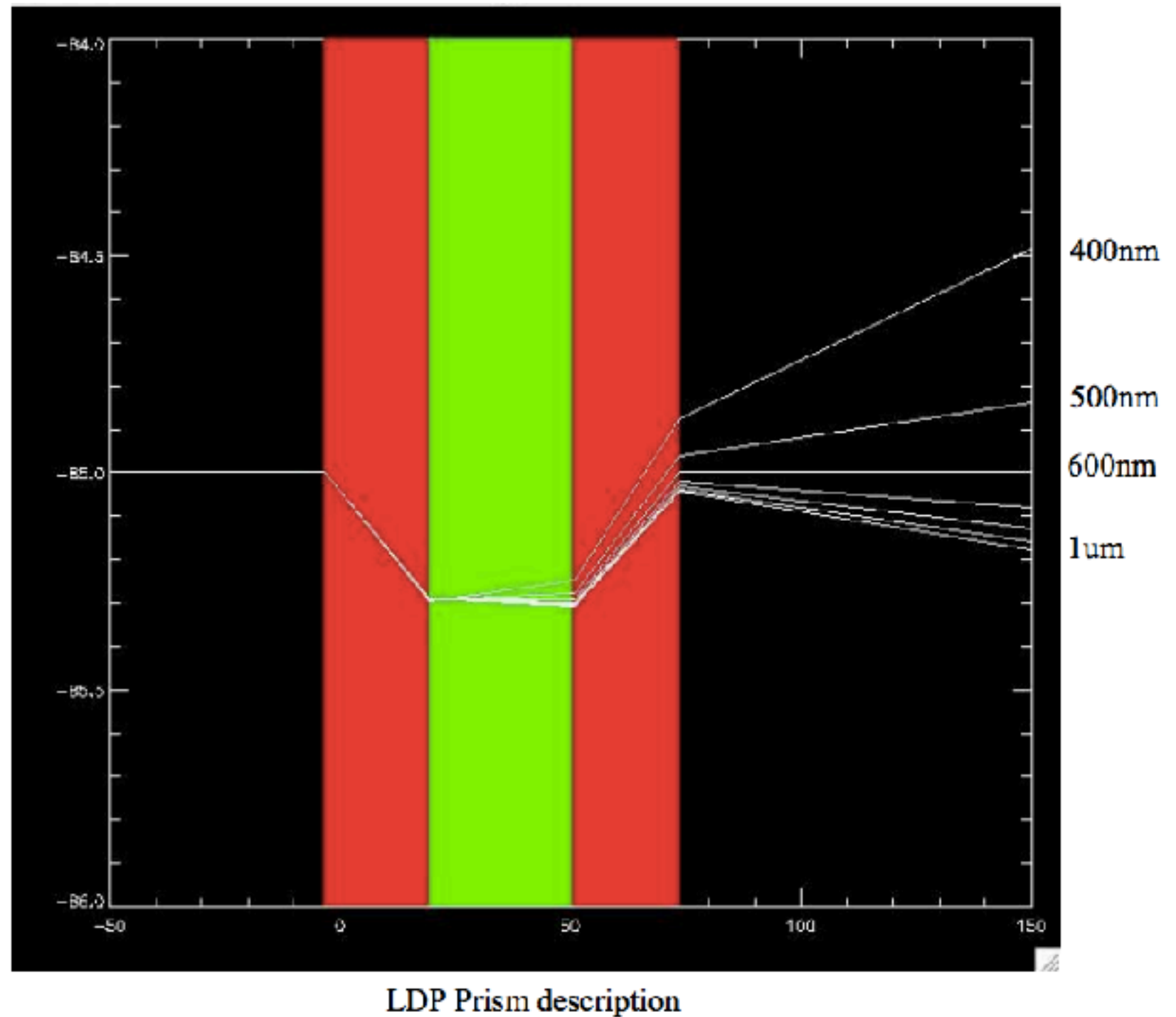


Olympus

Prism Spectroscopy

only few
astronomical
spectrographs
use prisms

- low dispersion
(resolution)
- dispersion
varies with
wavelength



“low dispersion prism” for IMACS spectrograph on
Magellan 6-m telescope; uses 3 prisms

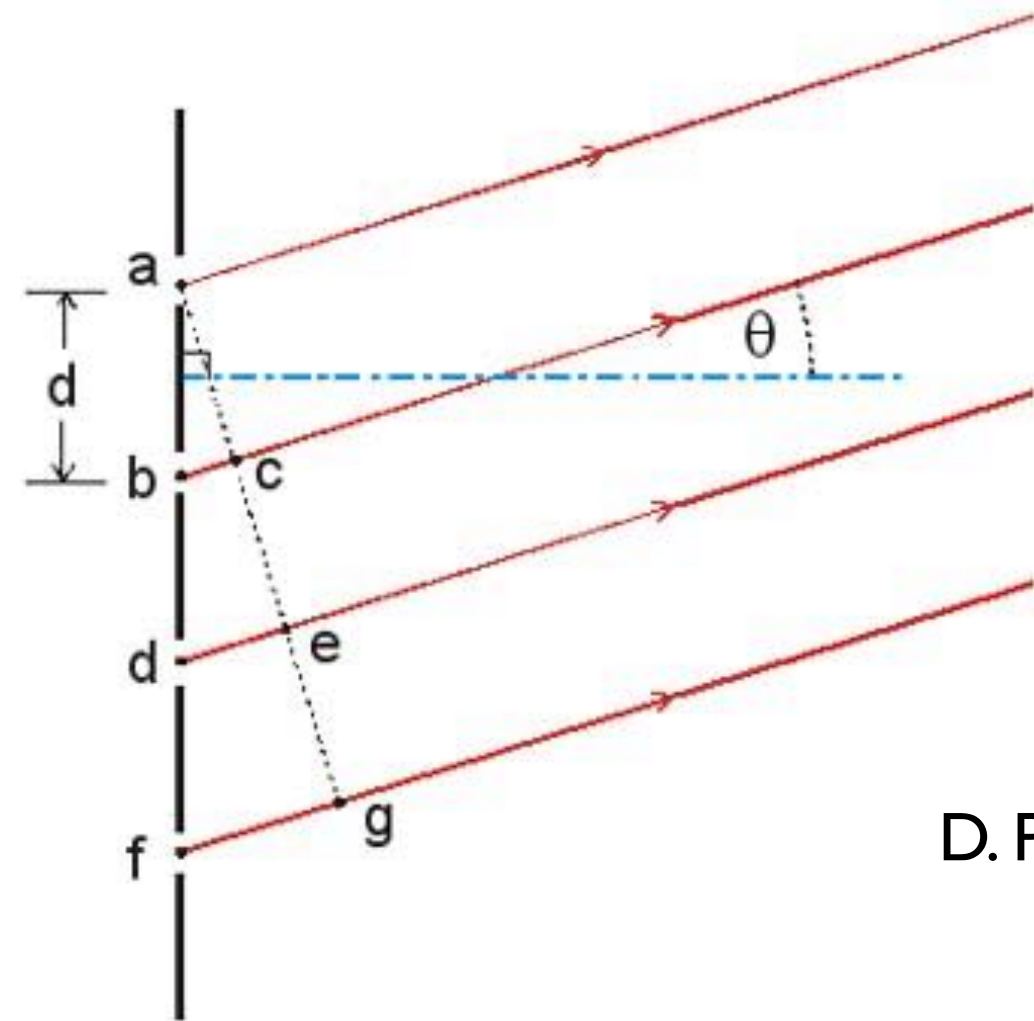
Diffraction gratings

make use of wave
properties of light:
interference

grating: many parallel
lines ($\sim 500/\text{mm}$)

similar to single-slit and
double-slit experiments

position of n th order:

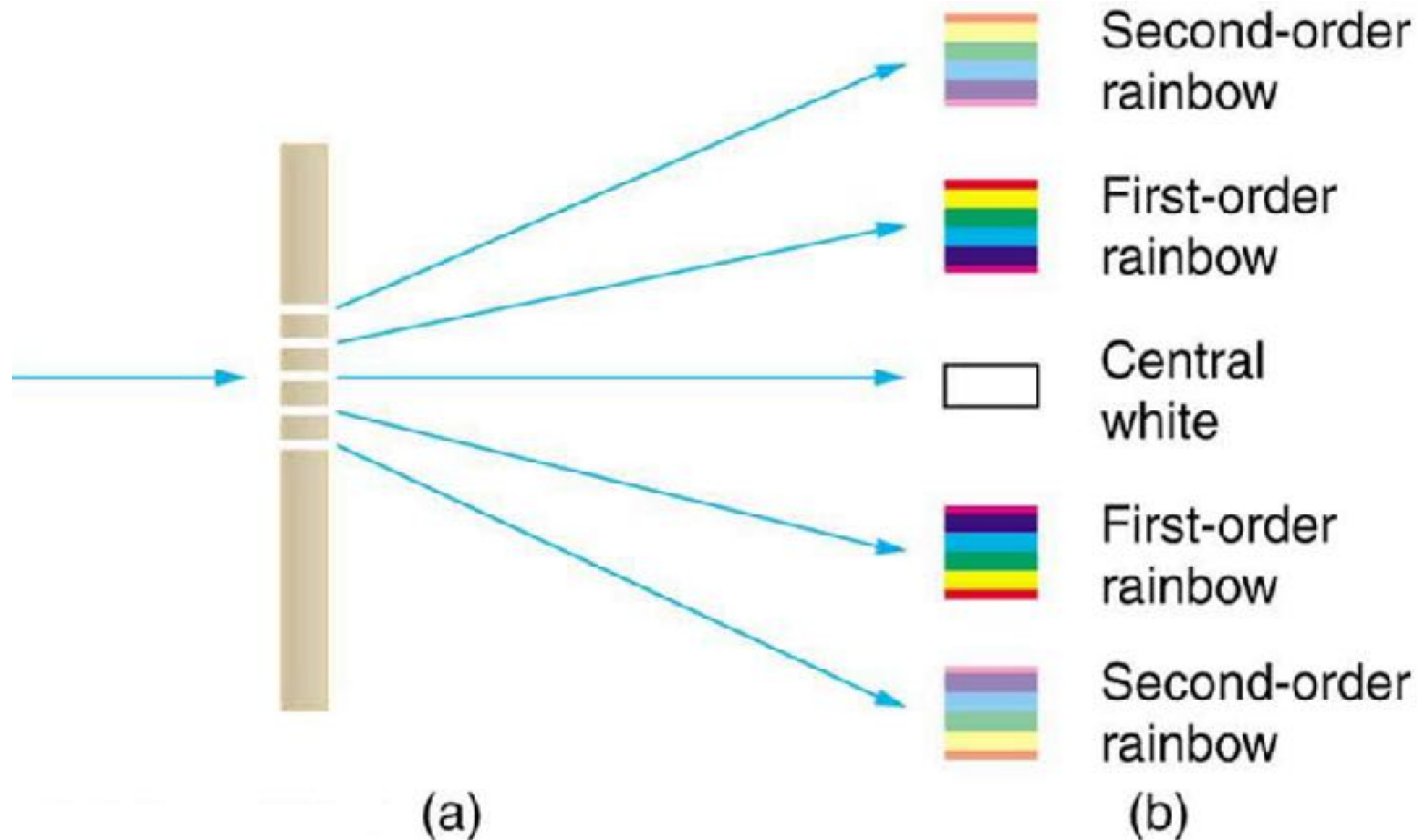


D. Pogosian

if $b-c = \lambda$: maximum at θ
and $d-e = 2\lambda$, etc.

$$n\lambda = d \sin \theta$$

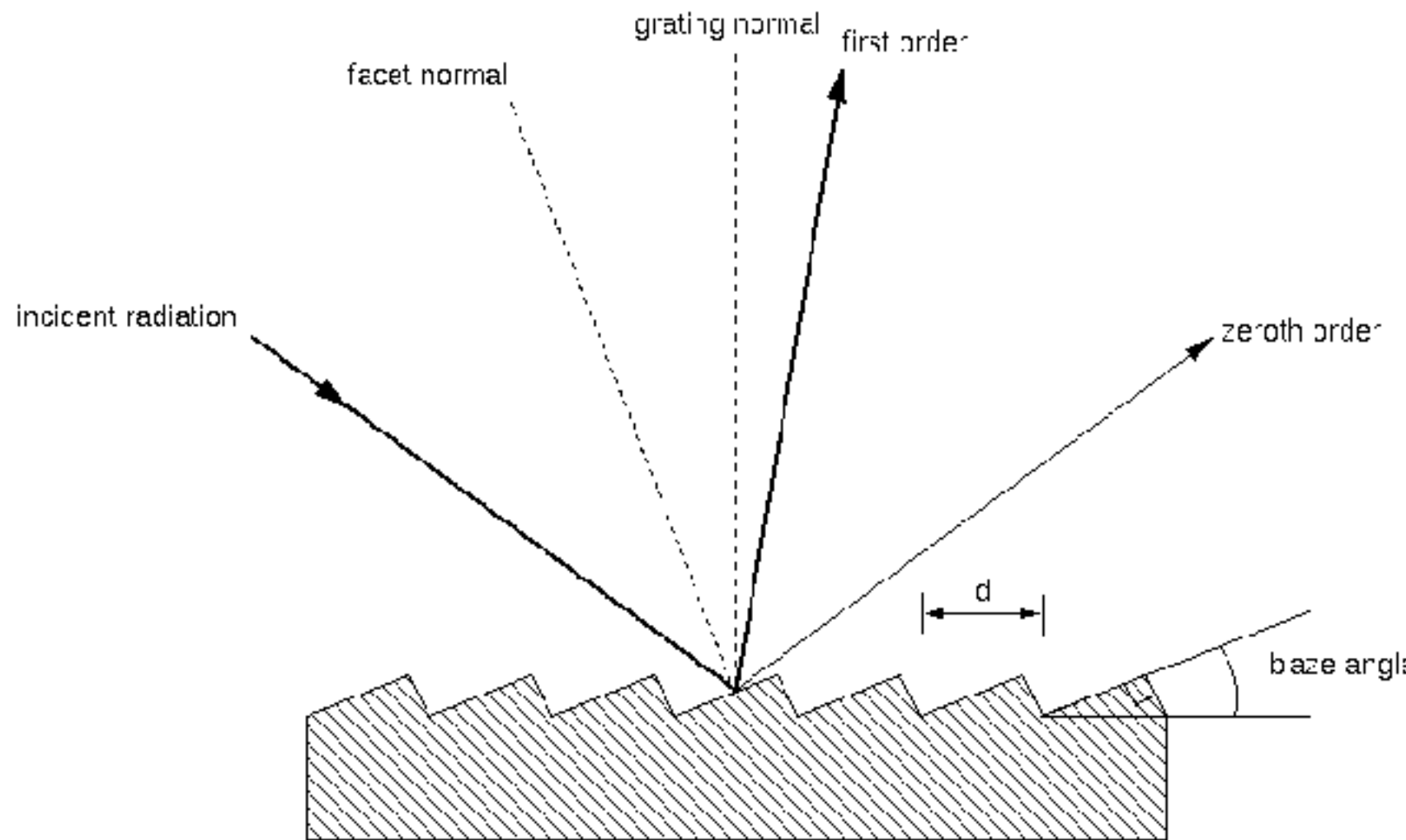
Diffraction gratings



Diffraction gratings

can be transmission gratings or reflection gratings

most astronomical spectrographs use reflection gratings



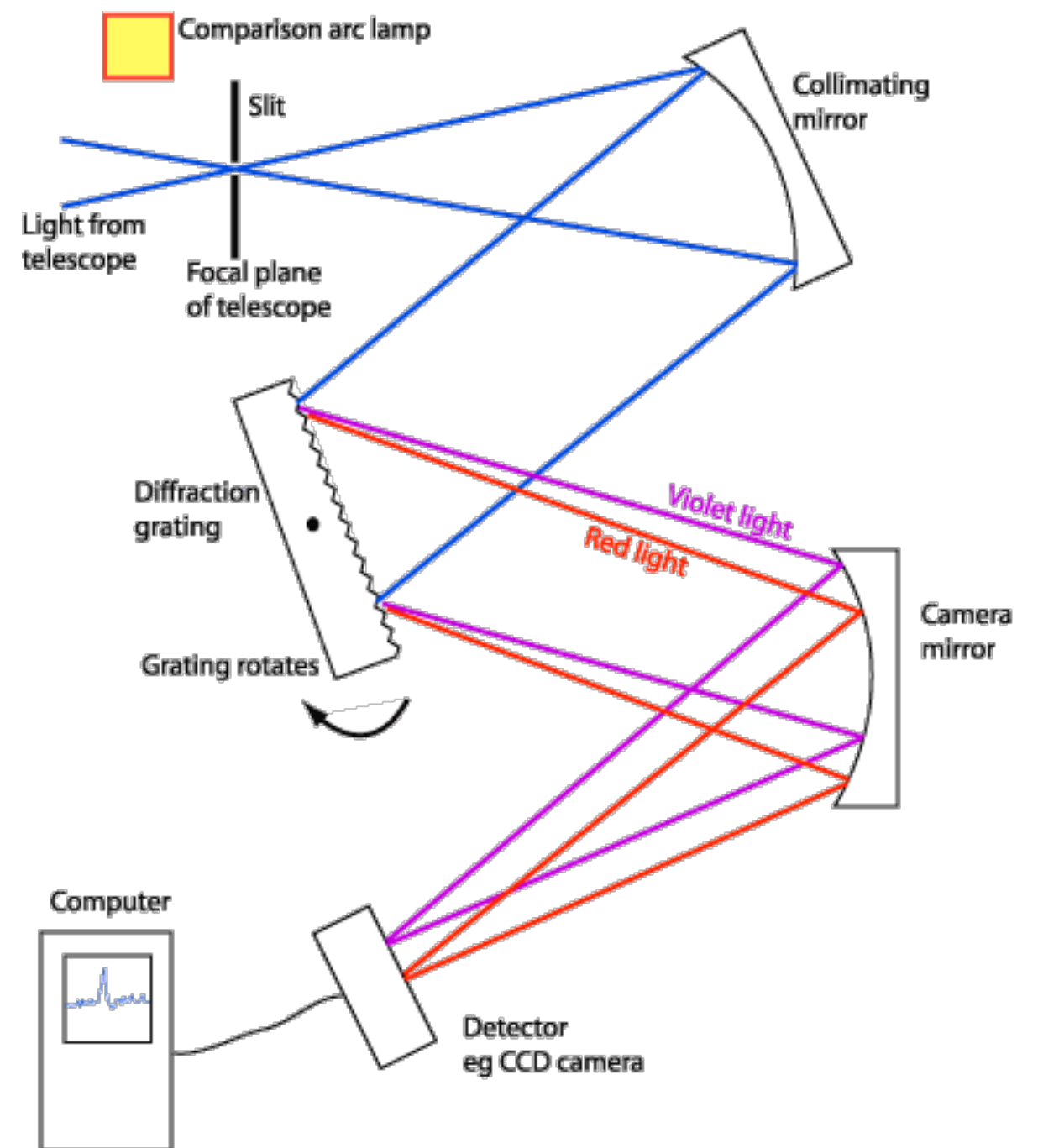
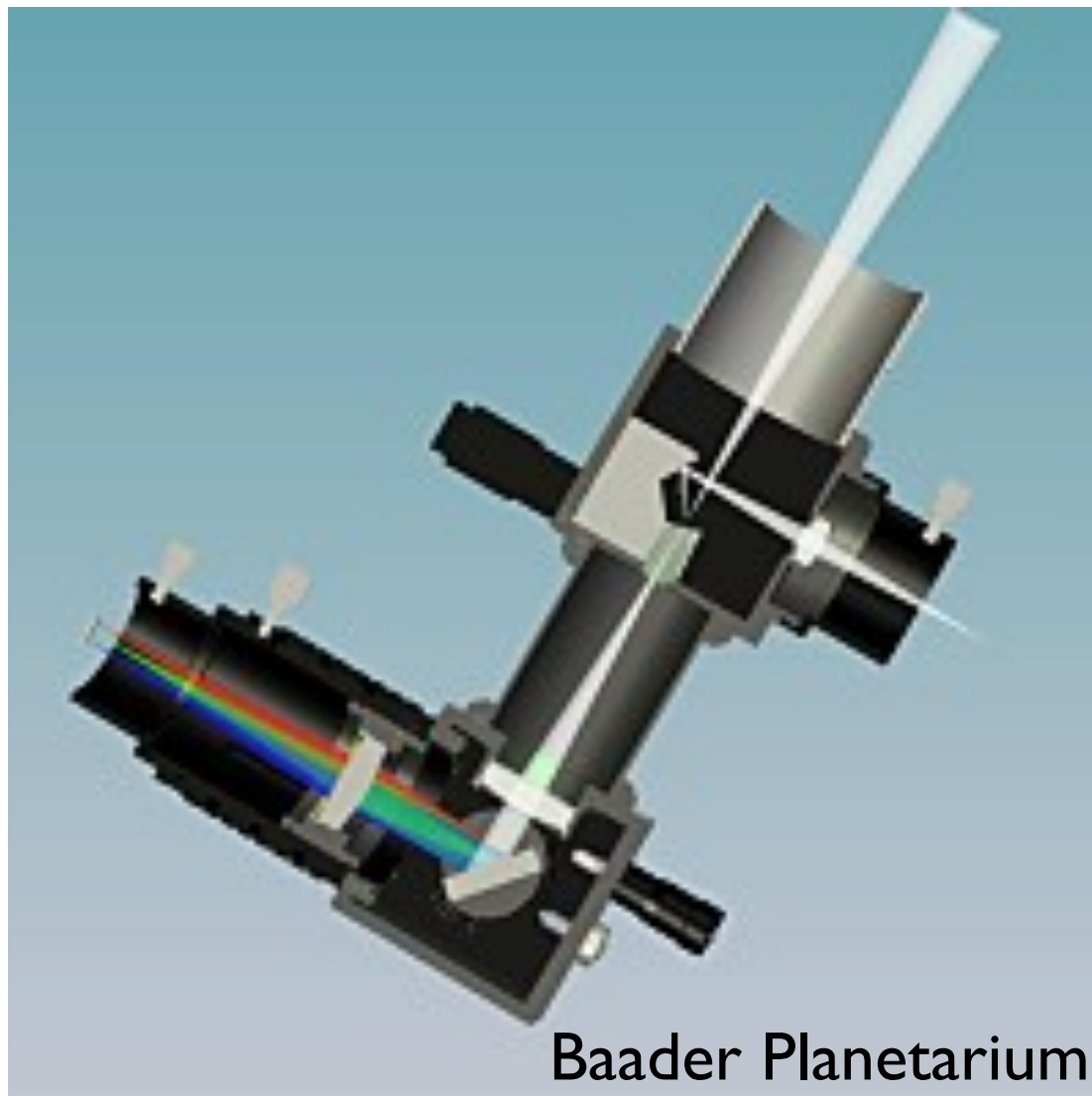
Dhillon 2012

blaze wavelength: wavelength for direction of reflection coincides with desired spectral order
→ maximal efficiency

Typical spectrograph

entrance: usually a slit, similar to seeing size

collimator: converts a diverging beam to a parallel beam

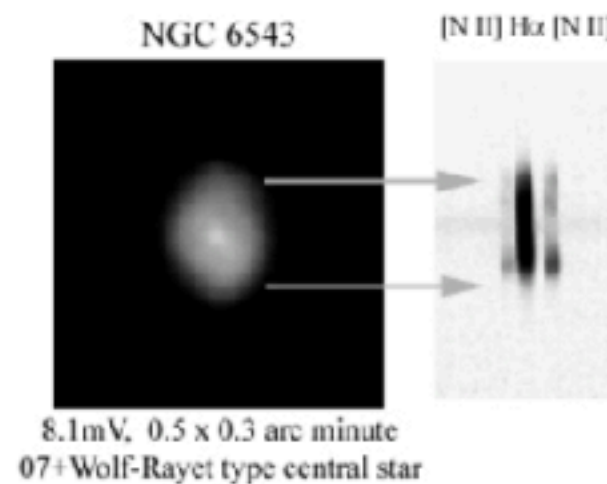
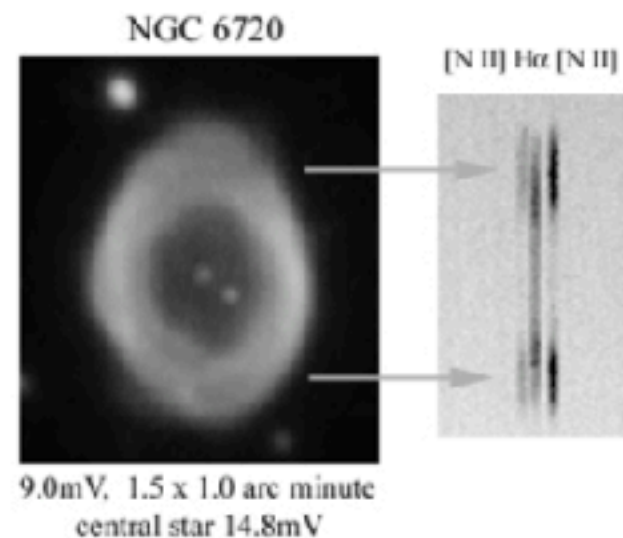
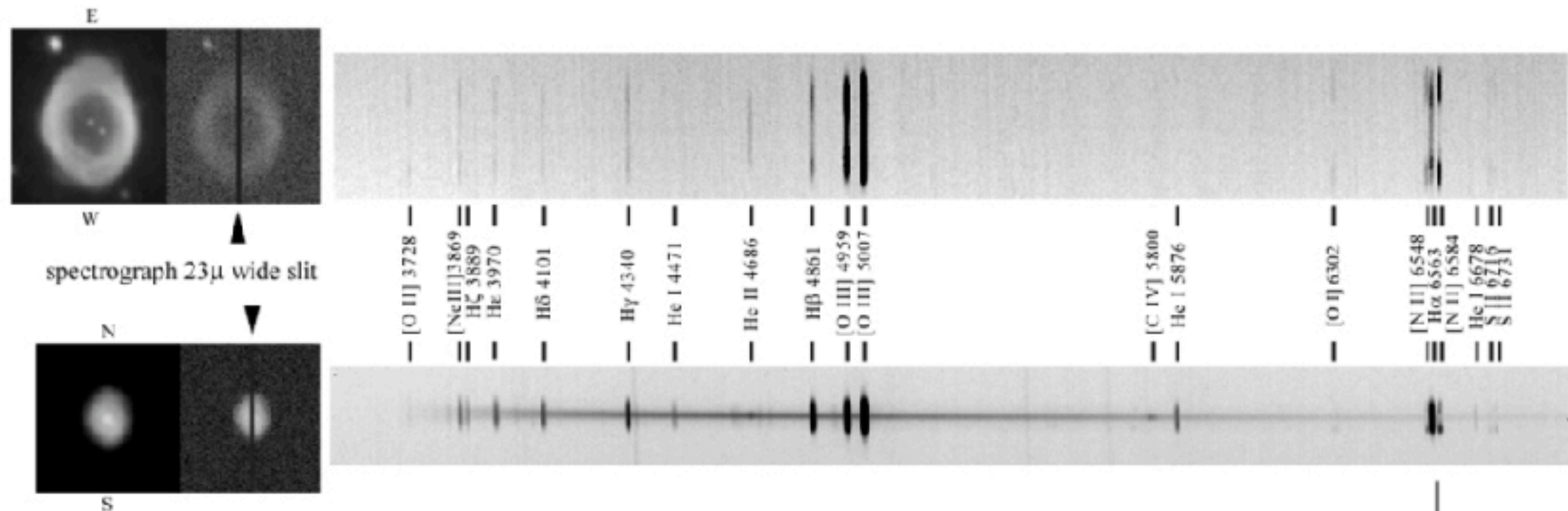


A Schematic Diagram of a Slit Spectrograph

Long-slit observations

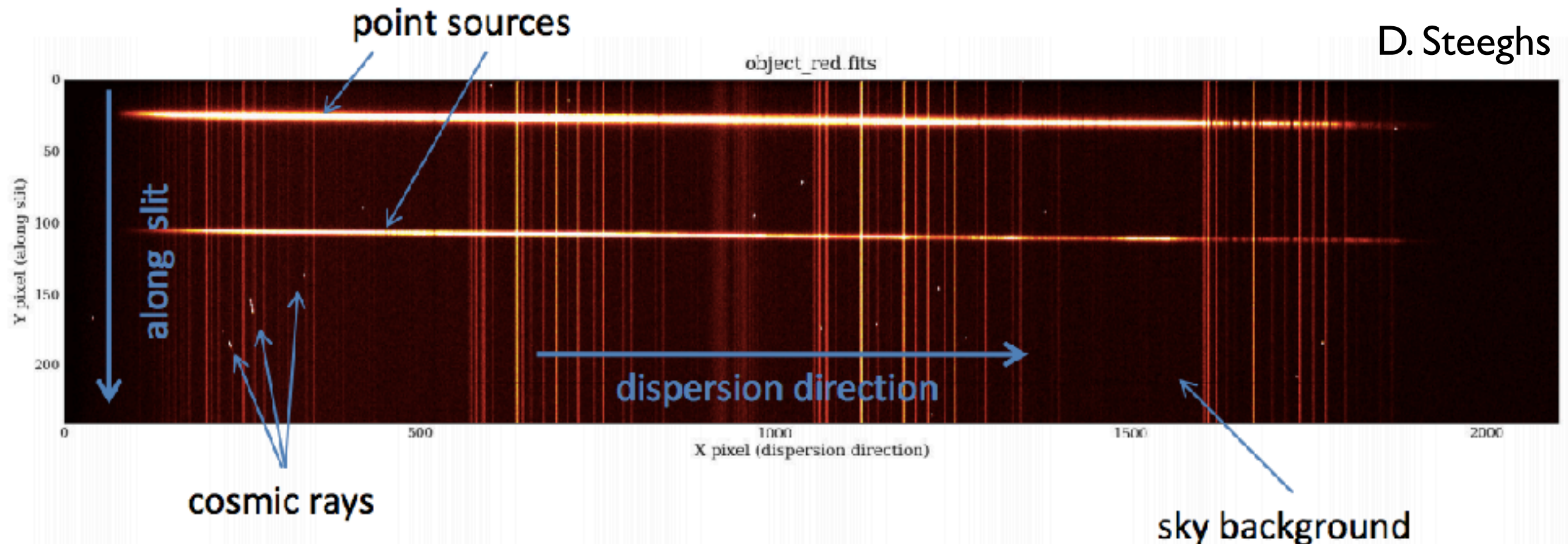
Planetary Nebula Spectroscopy : NGC 6720 [Ring Nebula] & NGC 6543 [Cat's Eye Nebula]

Jim Ferreira, Livermore CA



Long-slit observations

D. Steeghs



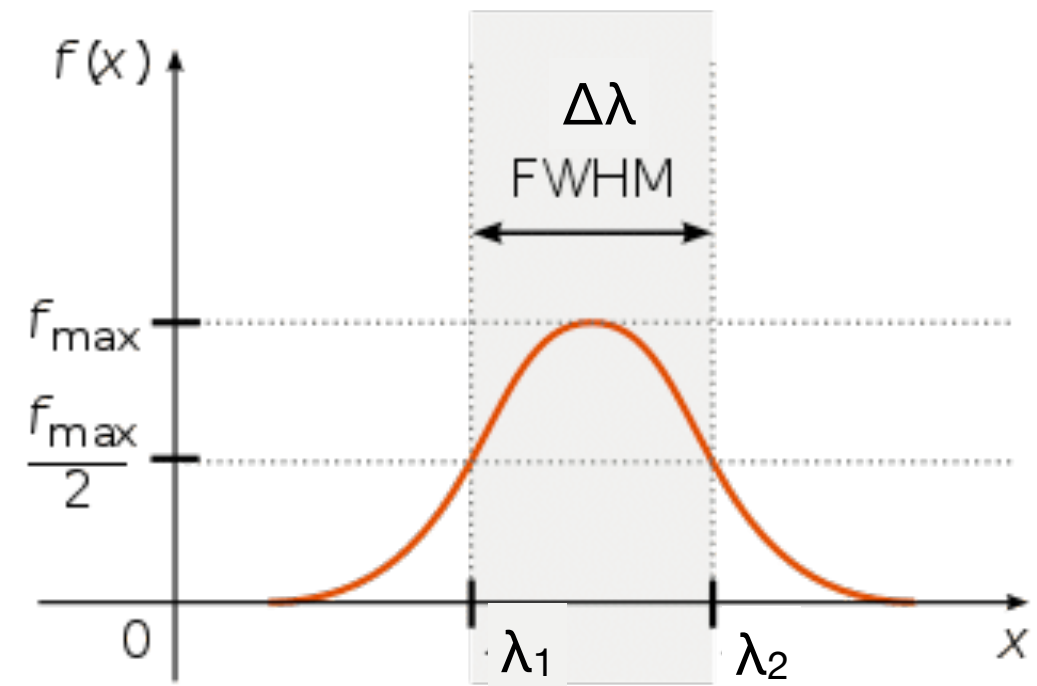
- long axis of CCD used to sample spectrum
- spatial information along slit still available: two objects, lots of sky
- sky background has a lot of emission lines!

Spectral Resolution

spectrographs have a finite resolution $\Delta\lambda$:

- FWHM of observed line width of incoming delta-function emission line
- minimum wavelength separation to resolve 2 lines

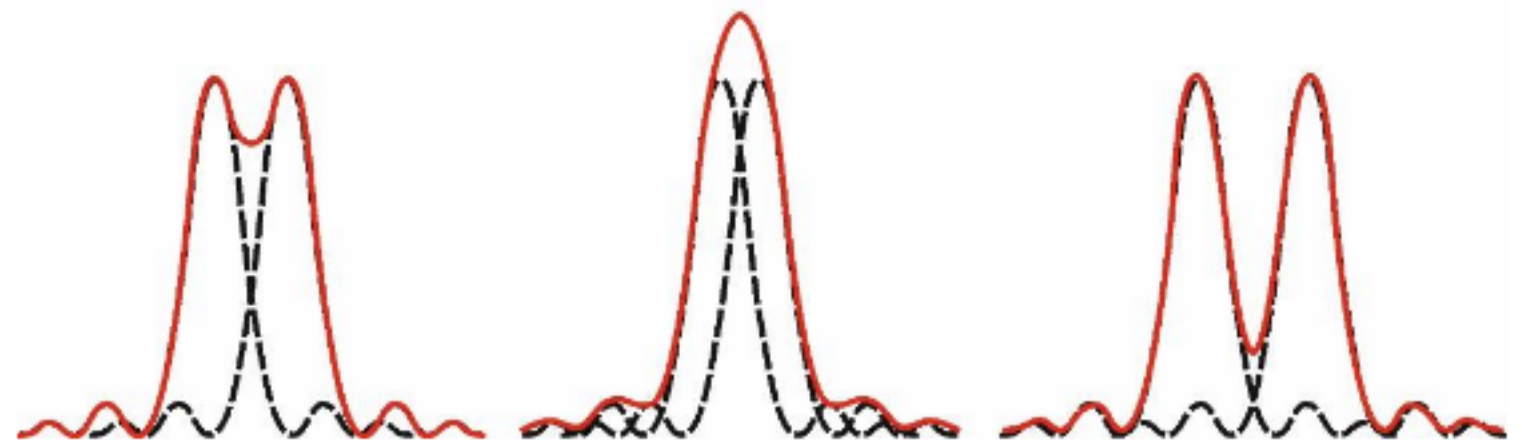
Note: very similar to the definition of *spatial* resolution of a telescope!



(a) Rayleigh criterion

(b) Unresolved

(c) Resolved



Spectral Resolution

often expressed as R :

$$R = \frac{\lambda}{\Delta\lambda}$$

determined by:

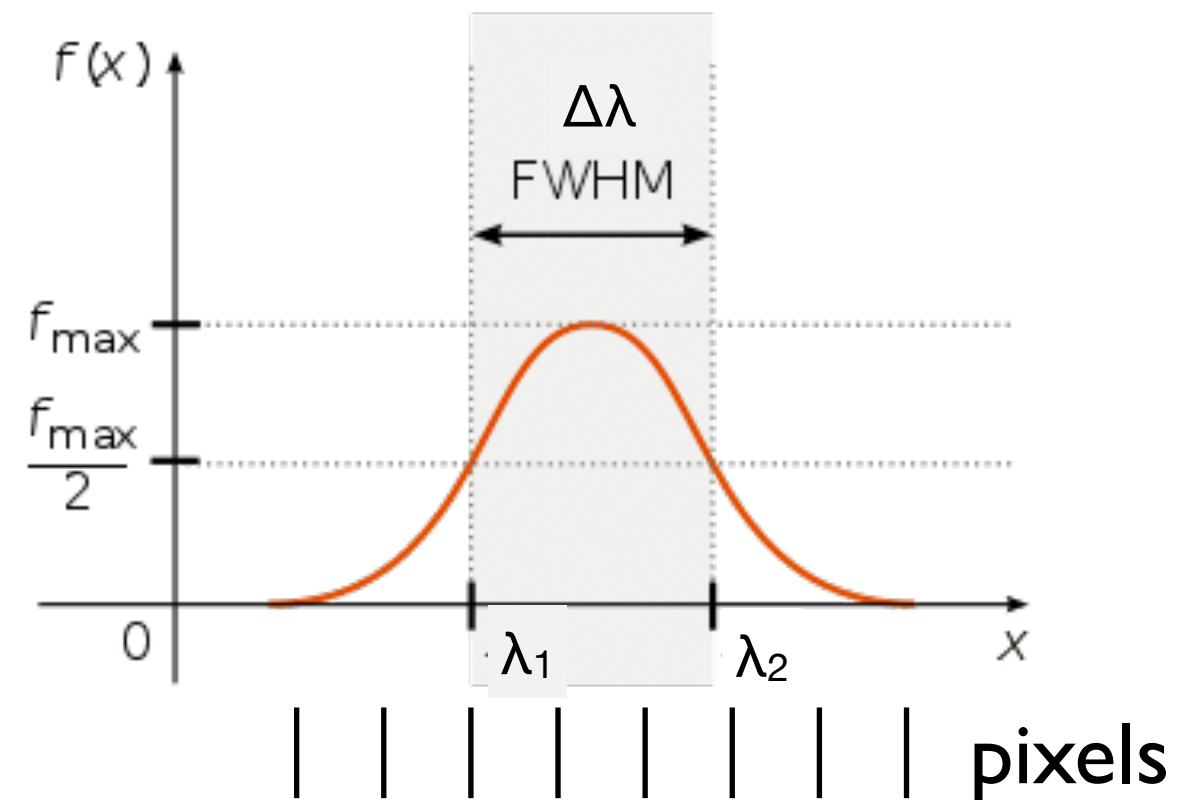
- grating (line density):
more lines $\rightarrow$ higher R
- width of entrance slit:
wider slit $\rightarrow$ lower R

$R < 1000$	low-resolution	e.g. our “low-resolution” spectrograph
$1000 < R < 10,000$	medium-resolution	e.g. our “high-resolution” spectrograph
$R > 10,000$	high-resolution	Echelle spectrographs

Spectral Dispersion

resolution: R or $\Delta\lambda$

dispersion: length $\Delta\lambda'$ of spectrum over single pixel, $[\text{\AA}/\text{px}]$



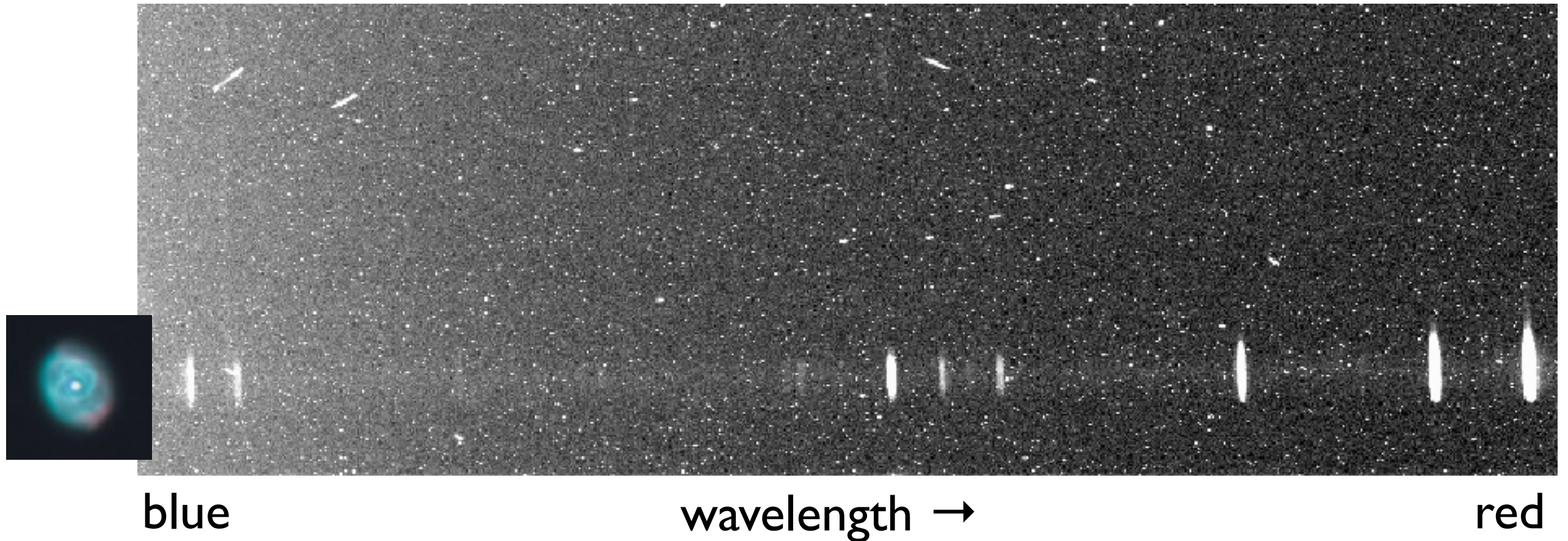
to properly sample the spectrum: $\Delta\lambda \sim 2 - 3 \Delta\lambda'$

analogous to imaging: size of PSF vs. pixel scale

Spectroscopic Calibration

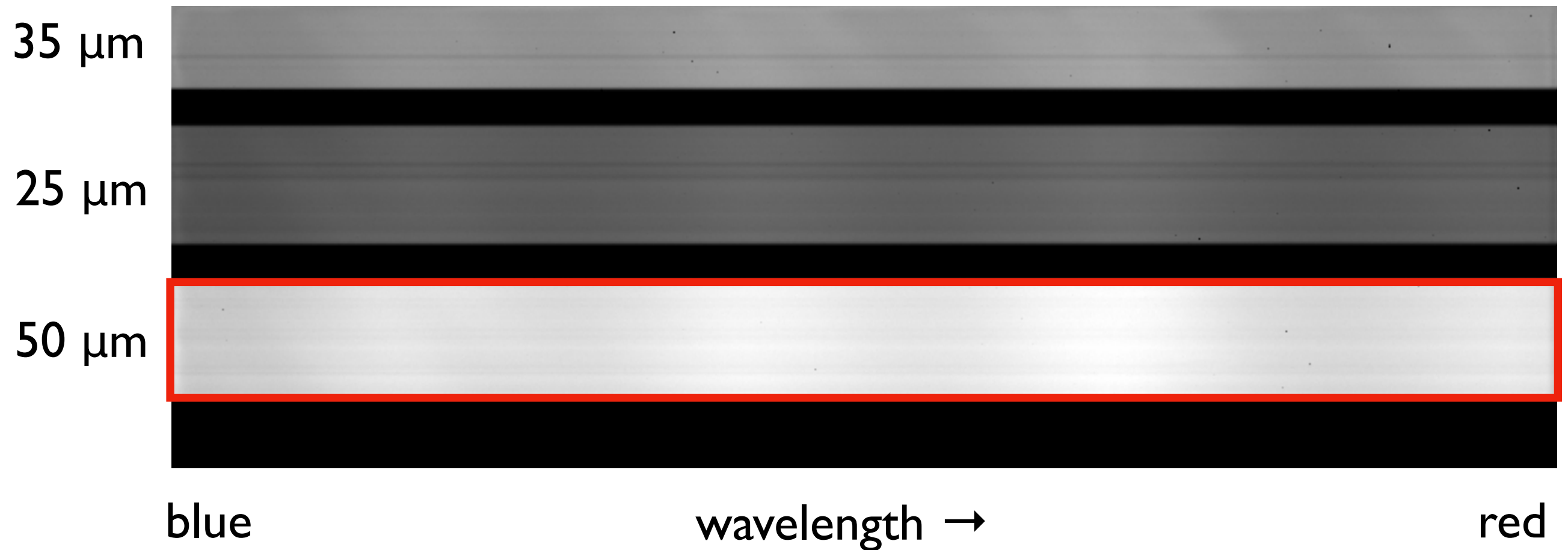
- dark frames!
- flat field: use bright continuum source
 - small-scale pixel sensitivity variation
 - variations in slit width
- wavelength calibration: which position on the CCD corresponds to which wavelength?
 - use “arc” lamps with discrete emission lines
 - can also use sky emission lines
- flux calibration:
 - “spectrophotometric” standard stars: stars with known spectral shapes, smooth continua

Spectroscopic Data Reduction



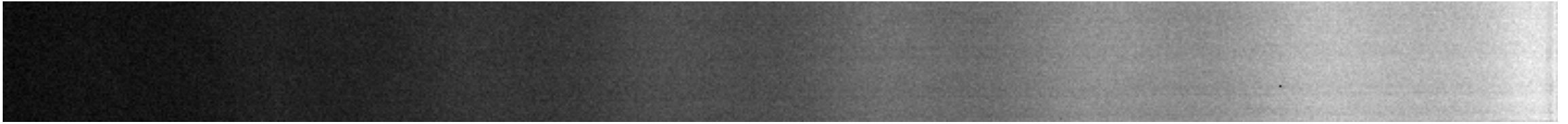
1. Subtract master dark
2. (Median-combine multiple exposures IF target always in the same place)

Spectroscopic Flat-field



3. Use flat-field to cut all images (flat, science, arcs) to desired pixel rows (only choose rows that are fully illuminated blue-red)

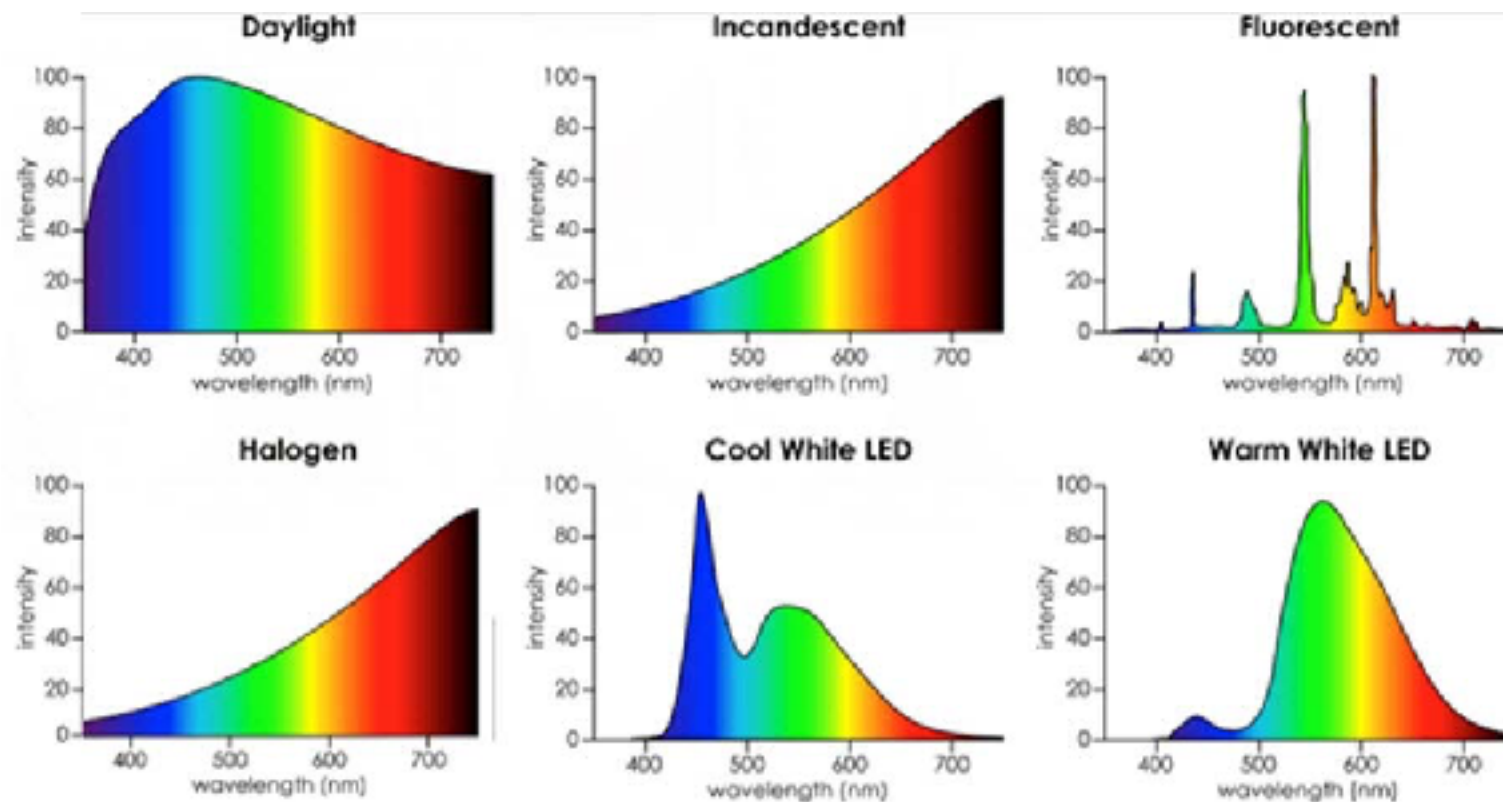
Spectroscopic Flat-field (2)



blue

wavelength →

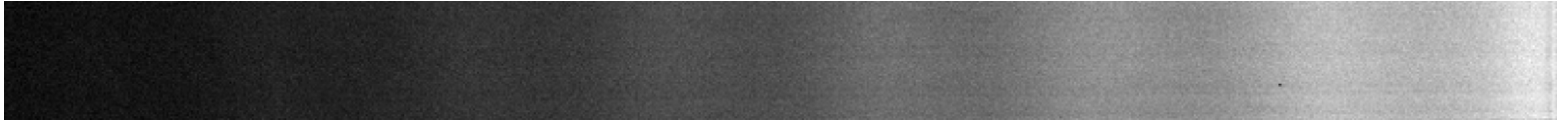
red



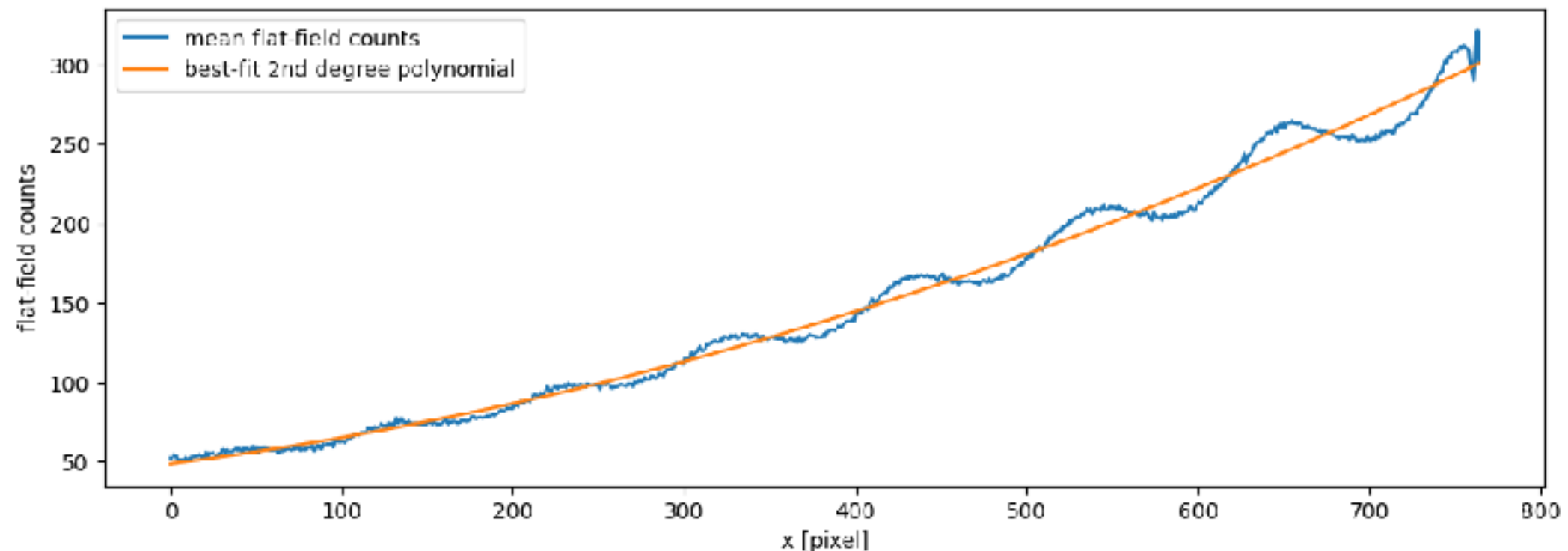
The overall shape of the spectroscopic flat-field is **not** indicative of sensitivity as fct. of wavelength.

Can still use it to correct pixel-to-pixel sensitivity variations.

4. “Collapse” 2-d flat-field to 1-d, by taking the mean (or median) of each column:



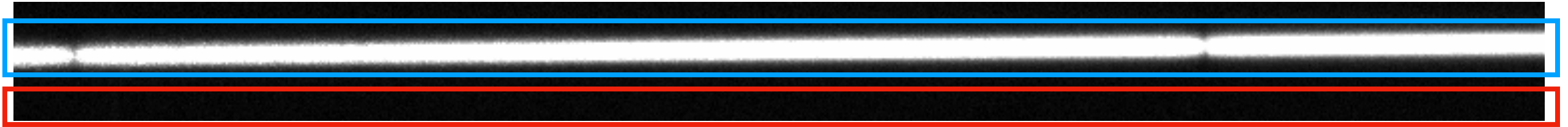
5. Fit low-order polynomial to 1-d flat-field:



6. Normalize 2-d flat by the best-fit polynomial - **THIS** is the spectroscopic flat-field; divide science frames by it



2-d $\rightarrow$ 1-d spectrum + background subtraction

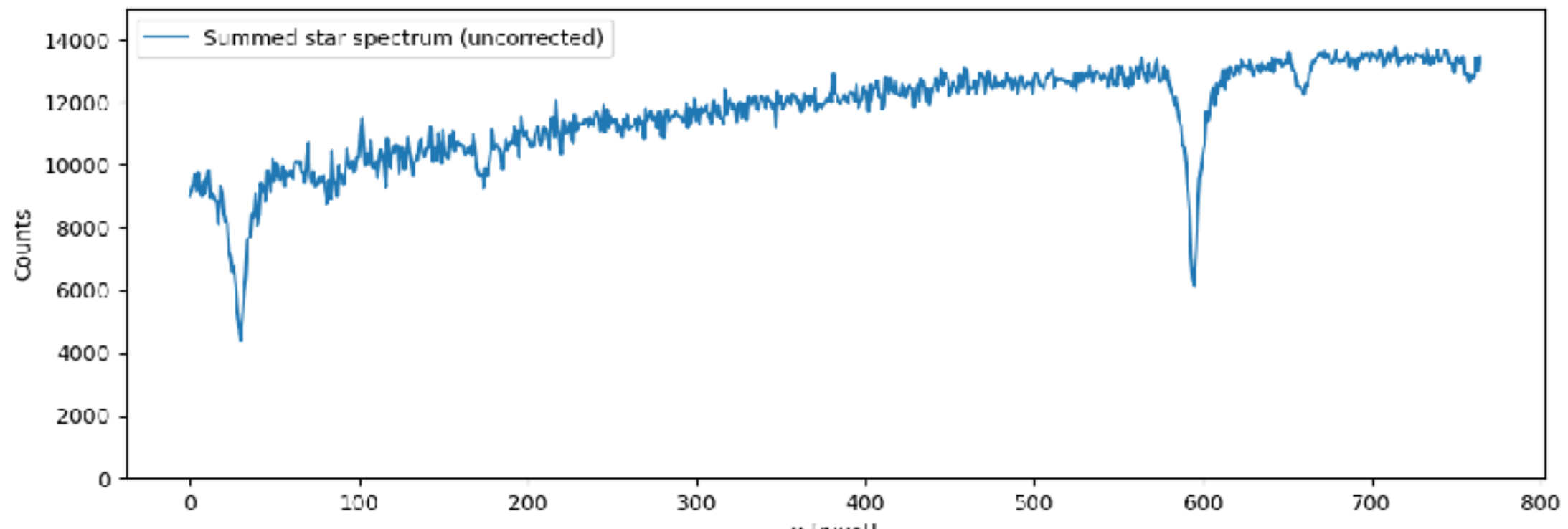


7. Define **source** rows: all rows with flux from object, and **background** rows: all empty rows.

Estimate *background flux per pixel* at given x-position (wavelength).

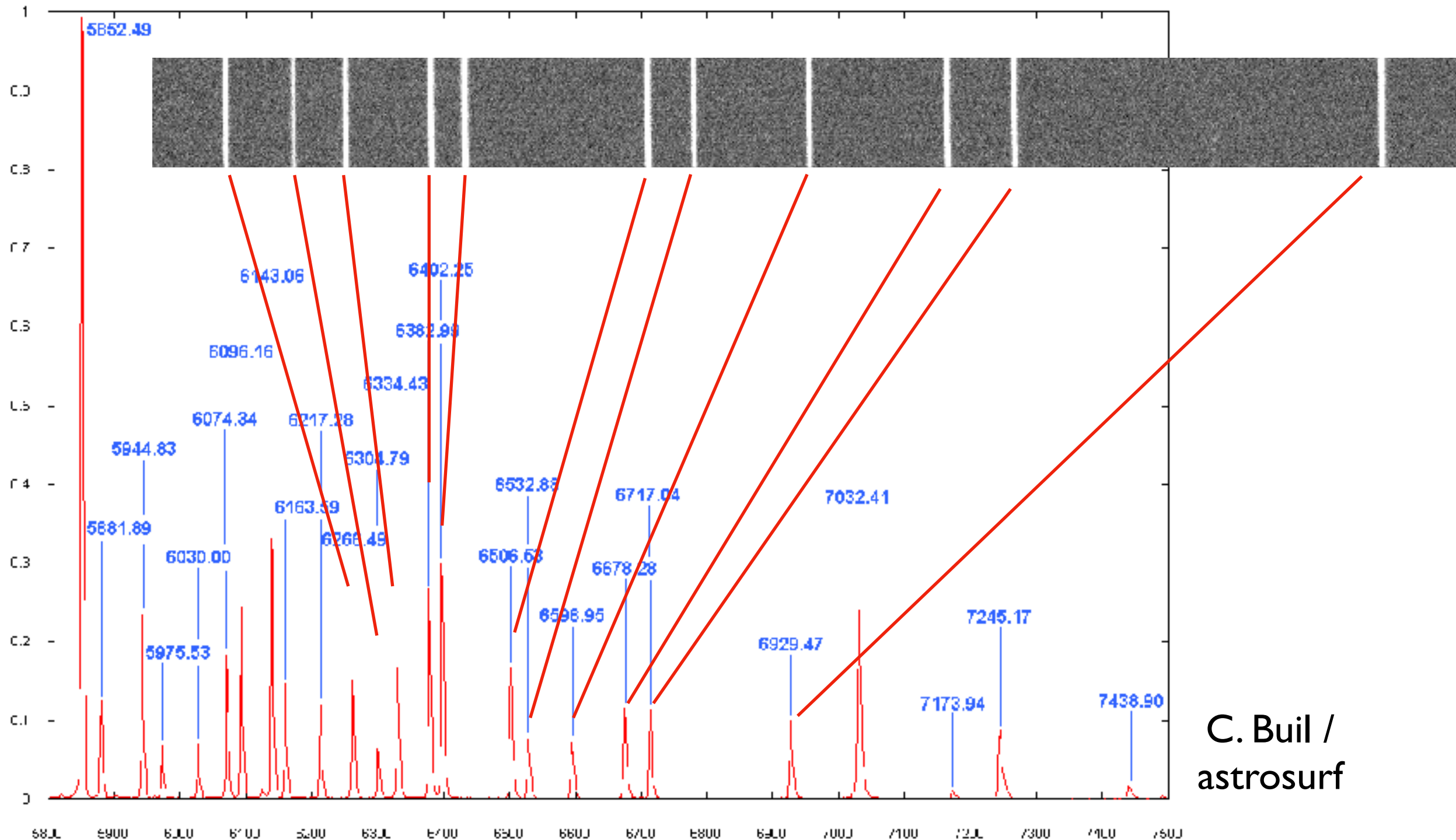
Subtract background from all rows.

Sum over source rows.



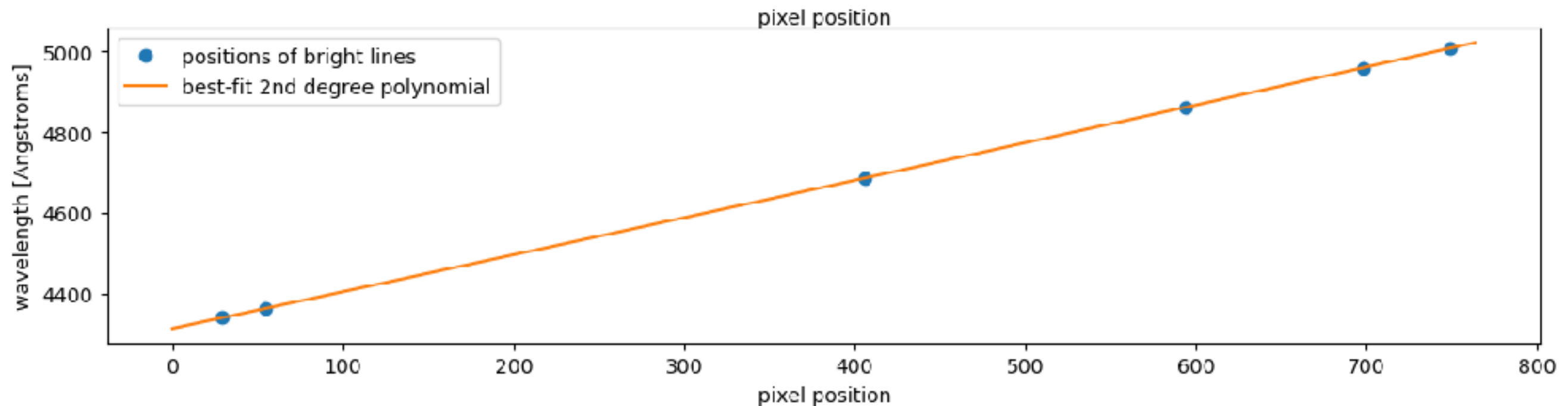
Wavelength Calibration

map pixel position to wavelength, using arc lamps (Ne, Hg, ...) or emission lines in science spectrum



Wavelength Calibration

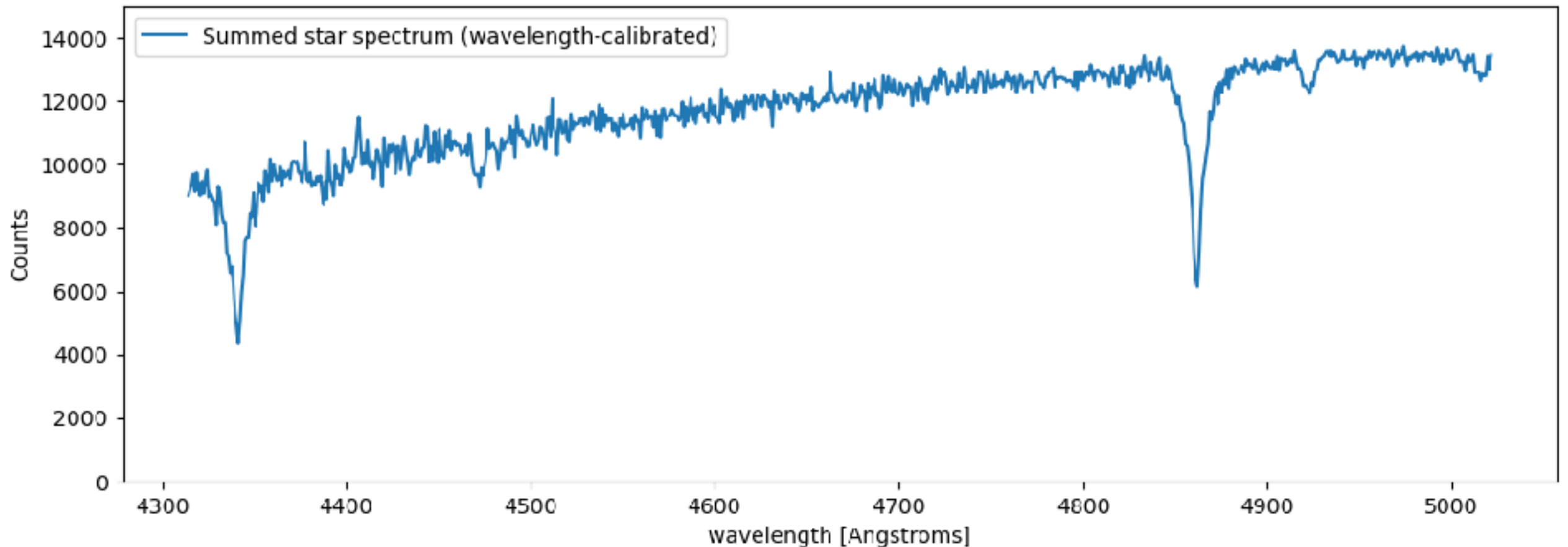
map pixel position to wavelength, using arc lamps (Ne, Hg, ...) or emission lines in science spectrum



8. fit wavelength as function of pixel position → wavelength calibration

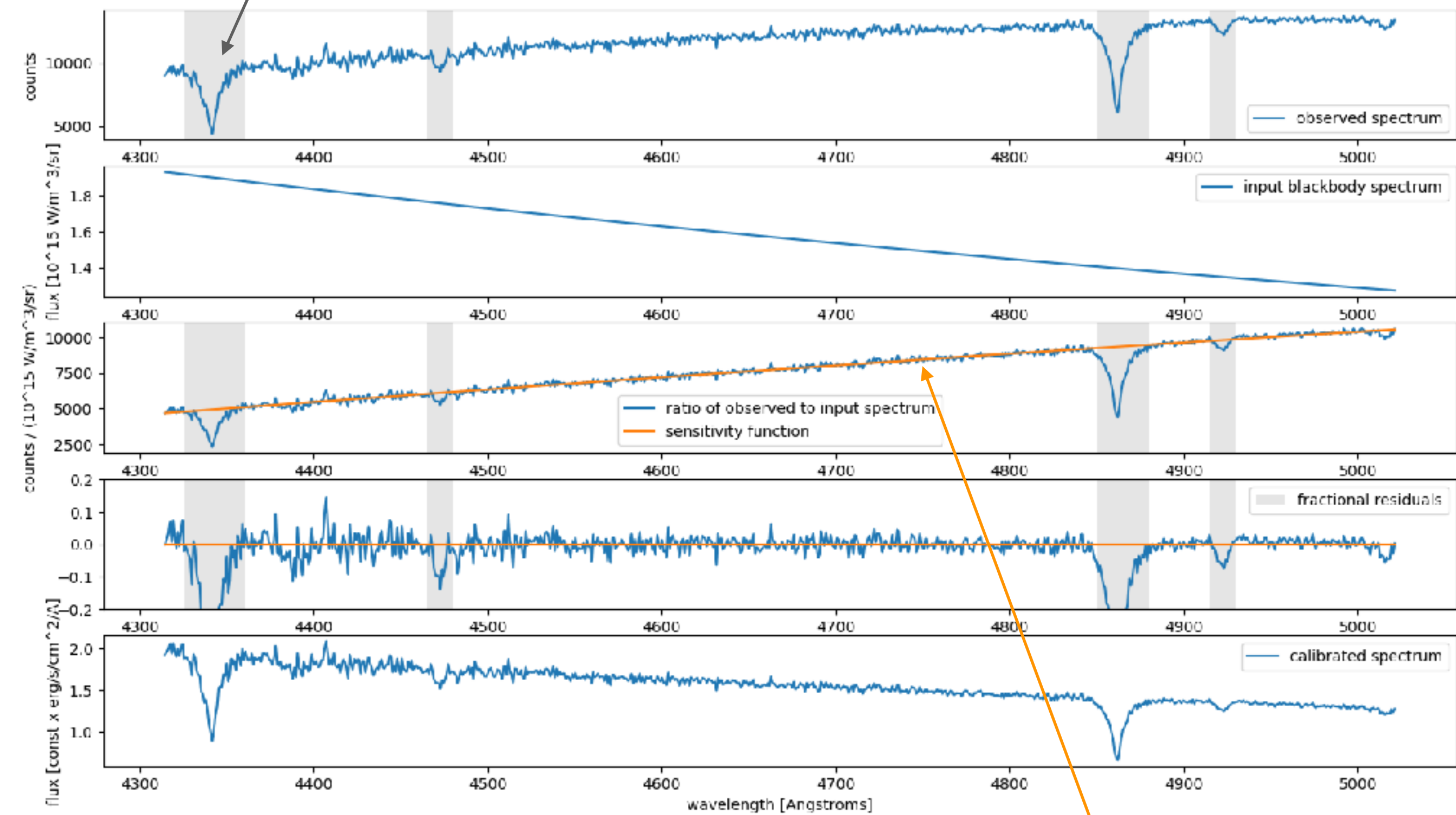
Flux Calibration

resulting spectra are in units of ADU counts, not flux.



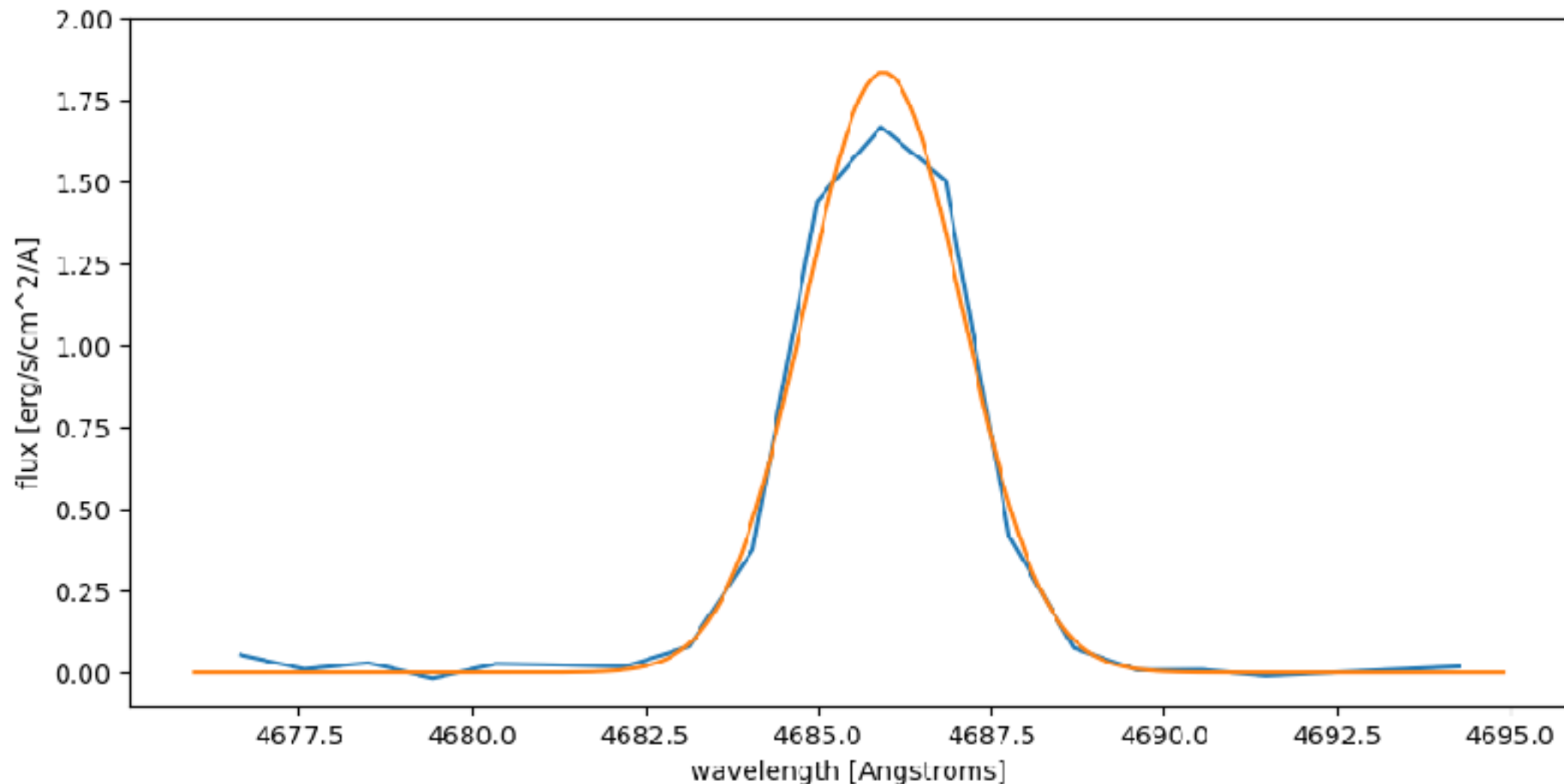
9. use observations of a star with known true spectrum to derive sensitivity function

masked areas



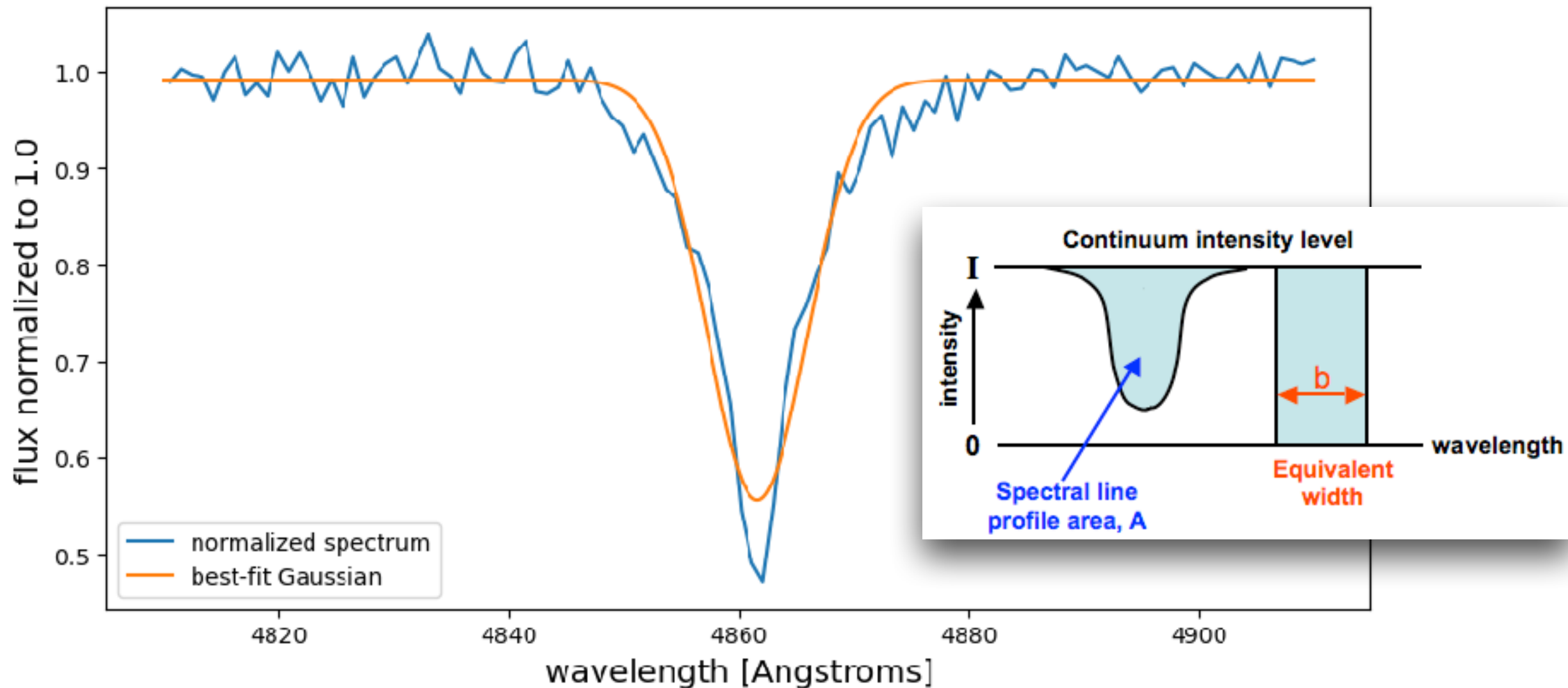
sensitivity function

Emission line measurements



- generally, spectrum has to be flux-calibrated
- fit Gaussian to emission line
 - **line strength:** area of Gaussian, units erg/s/cm^2 (or: subtract background, integrate the measured line flux)
 - **line width:** σ (or FWHM) of Gaussian, units $\text{\AA}$ or km/s

Absorption line measurements



- first, normalize by continuum
- spectrum does not need to be flux-calibrated
- fit Gaussian* to absorption line
 - **line strength** = “**equivalent width**”: area of Gaussian, units $\text{\AA}$
 - **line width**: σ (or FWHM) of Gaussian, units $\text{\AA}$ or km/s

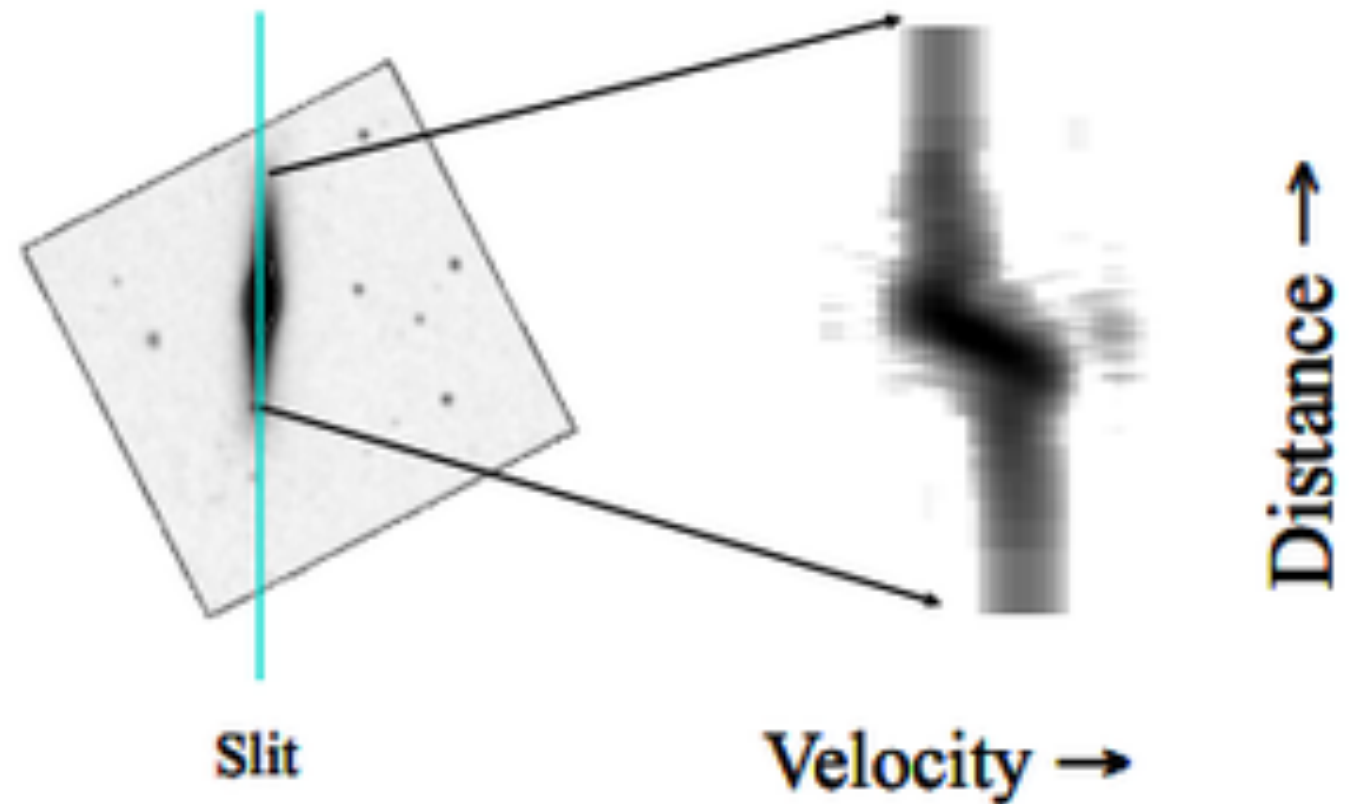
Long-slit spectrographs

most common spectrograph

can only target one (or a few) objects

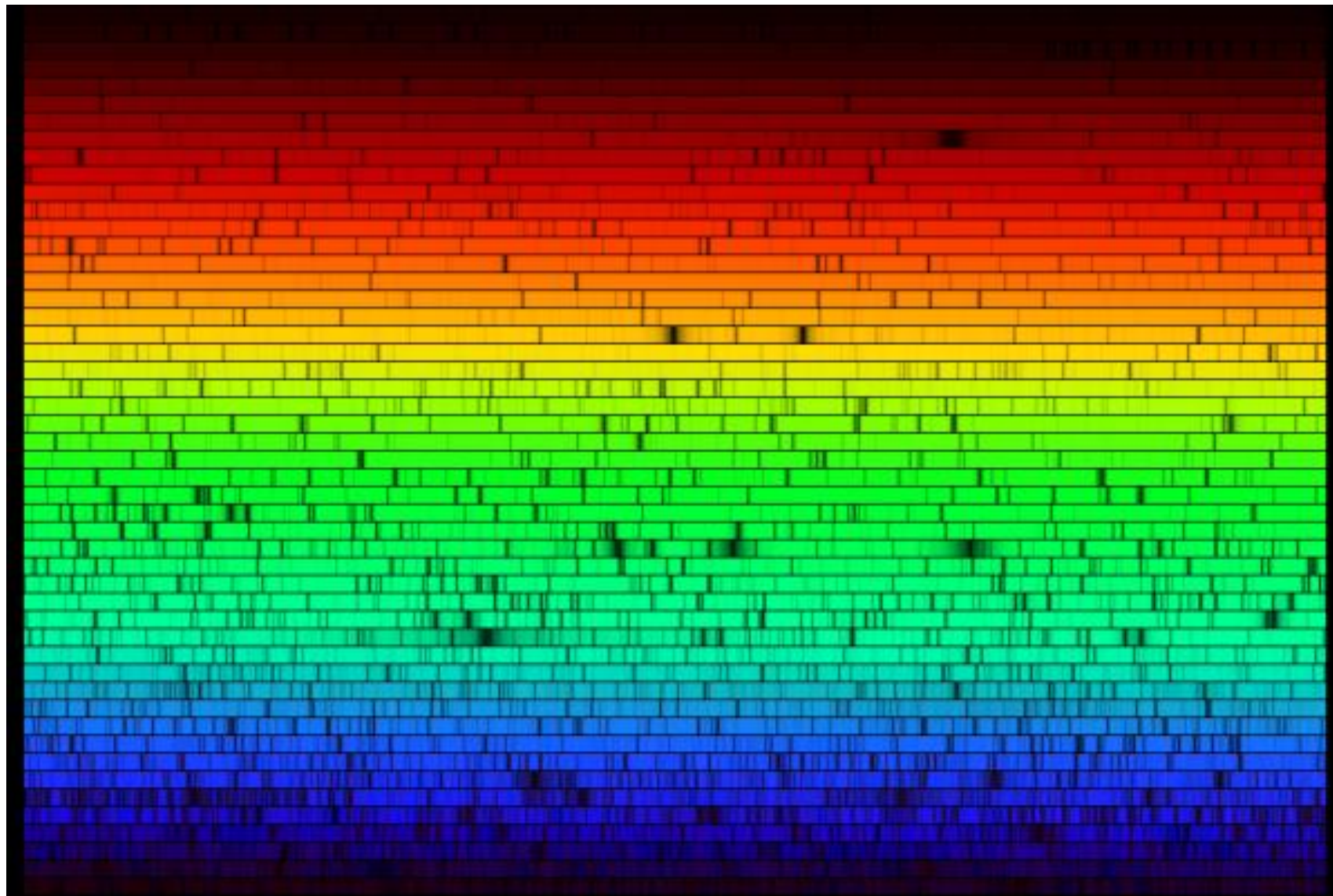
gives spatial variation

very good estimate of sky background



Echelle spectrographs

- very high resolution long-slit spectrographs
- have additional elements to fit entire spectrum onto CCD
- only for bright objects

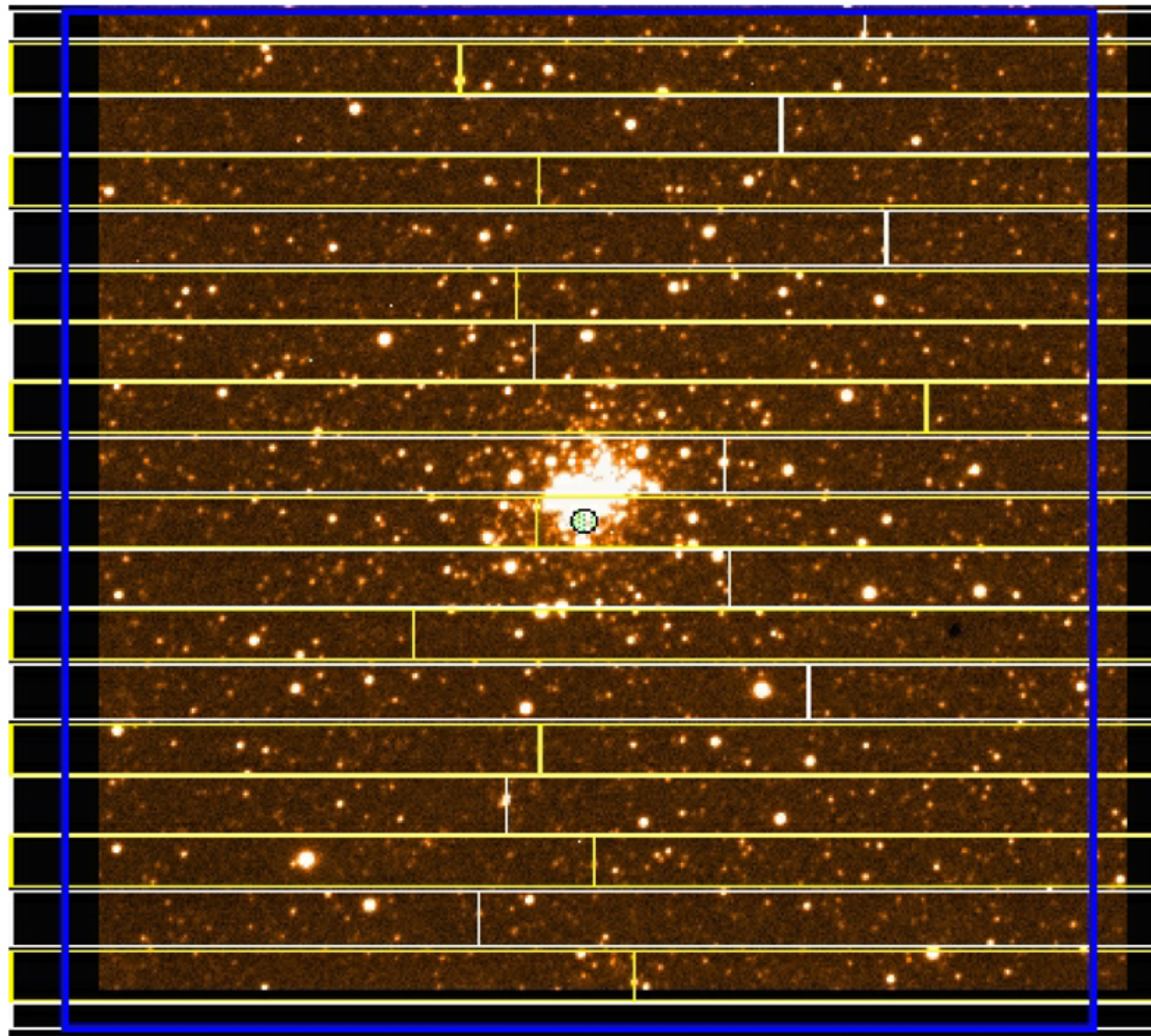


Echelle spectrum
of the Sun,
4000-7000Å

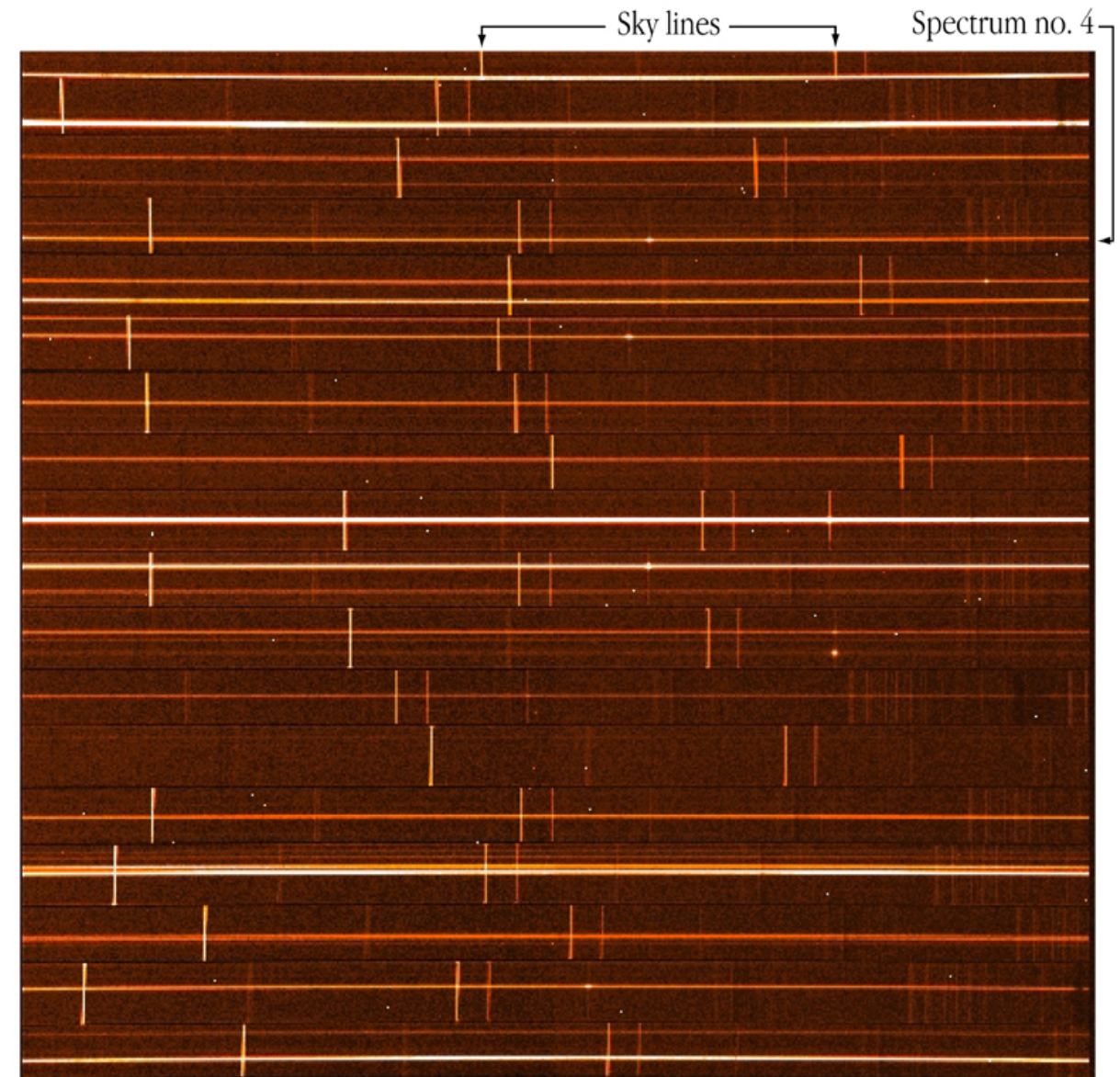
NOAO

Multi-object spectrographs

make a mask with multiple slits, one per target



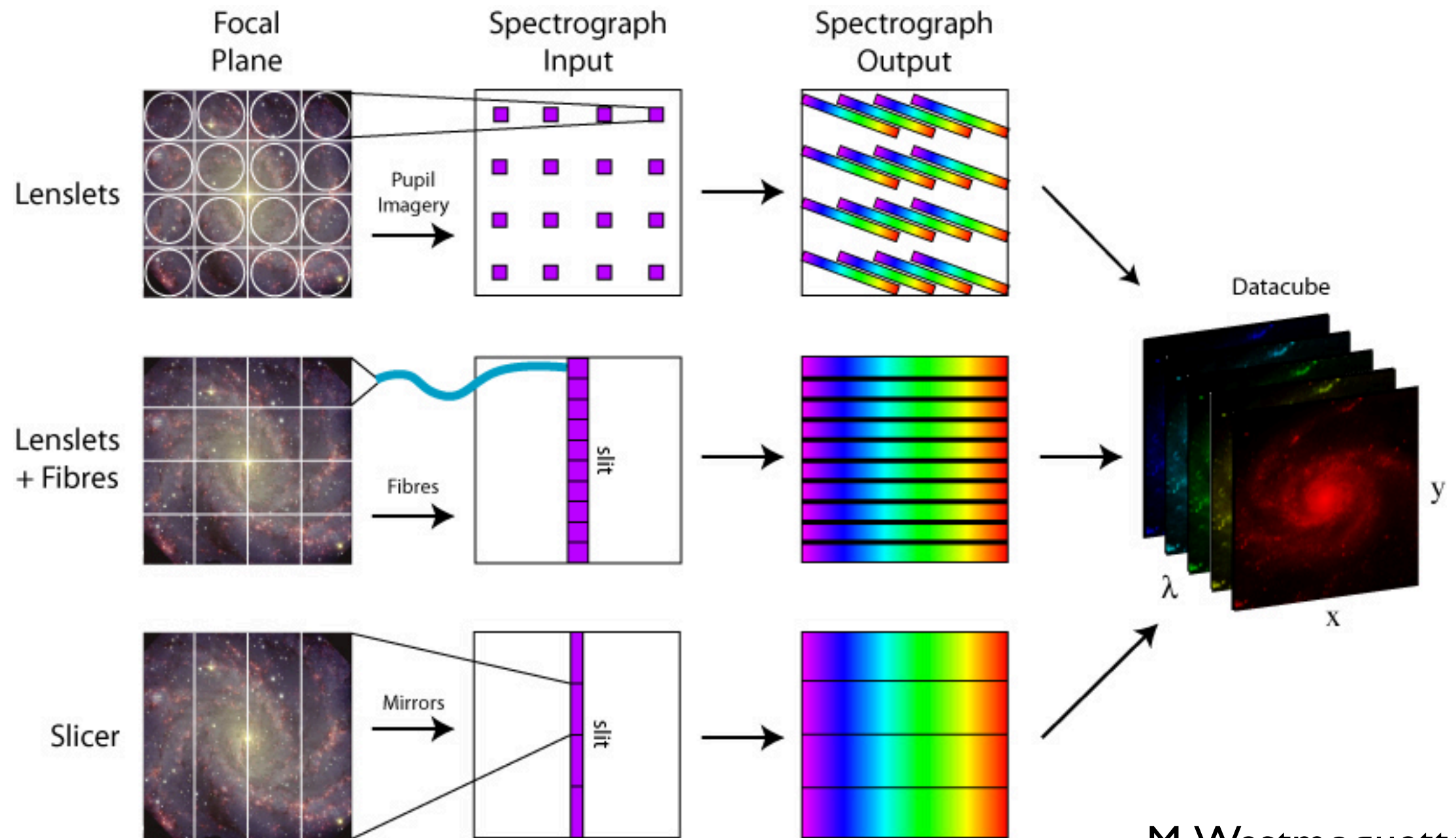
Open Cluster NGC 330 in SMC - VLT UT1 + FORS1 (MOS-mode)



Spectra of Stars in Open Cluster NGC 330 in SMC - VLT UT1 + FORS1 (MOS-mode)

Integral-Field Units

divide image into “spaxels” (spectroscopic pixels)



M. Westmoquette

Today's Tutorial(s)

- finish last week's tutorials
- spectroscopy notebooks
- photometric measurement tutorial

Spectroscopy

anjavdl edited this page 13 minutes ago · [12 revisions](#)

Edit

New page

Data acquisition:

[General guide](#)

Data reduction:

[General guide](#)

Lab options

1. [Spectroscopy of an Emission-Line Nebula](#)
2. [Spectroscopy of Bright Stars](#)

Resources for the data analysis:

[Some helpful python commands](#), reading in [this file](#) and [this file](#).
Tutorial on [Spectrophotometric flux calibration](#).

Resources for the observations:

Equipment quick-start instructions and manuals
(also see [Observing Equipment](#)):

[Telescope Quick-Start](#)
[Spectrograph Quick-Start](#)
[Spectrograph Manual](#)
[CCDSOFT Quick-Start](#)
[AutoGuider Quick-Start](#)

Reference arc lamp spectra:

[Neon](#)
[Mercury](#)

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General Information

- [Syllabus](#)
- [Schedule w/ links to slides, HW, etc.](#)
- [Grading](#)
- [Academic Policies](#)

Labs and Write-Ups

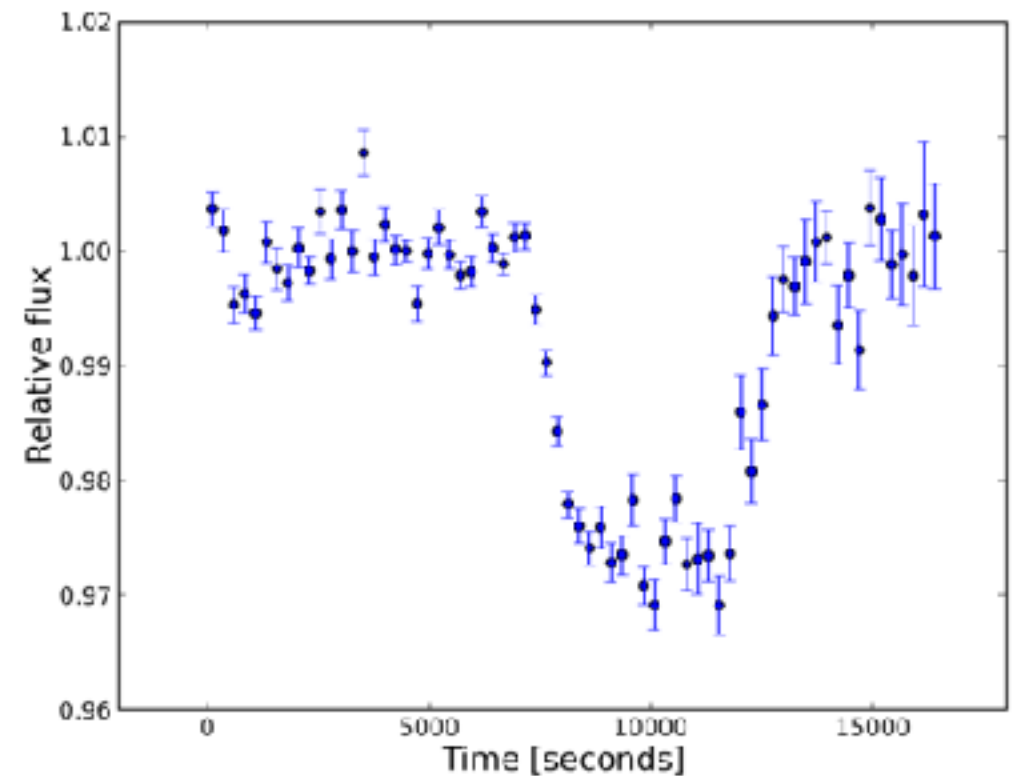
- [Lab report guidelines](#)
- [Observing Equipment](#)
- [Observing Calendar](#)
- [Lab 1: CCDs](#)
- [Lab 2a: Exoplanet transit](#)
- [Lab 2b: Spectroscopy](#)
- [Lab 3: Your own proposal](#)
- [Discontinued: Radio Interferometry](#)
- [Astronomical Data Archives](#)
- [Weather](#)
- [End-of-night report](#)

Computing

- [Computing Resources](#)
- [Astro Software Overview](#)
- [Bash](#)
- [awk and sed](#)
- [LaTeX](#)
- [Python](#)
- [jupyter](#)
- [GitHub](#)
- [ds9](#)
- [SExtractor](#)
- [Topcat](#)
- [Astrometry.net](#)
- [dfits and fitsort](#)
- [Image arithmetic \(+ftools\)](#)

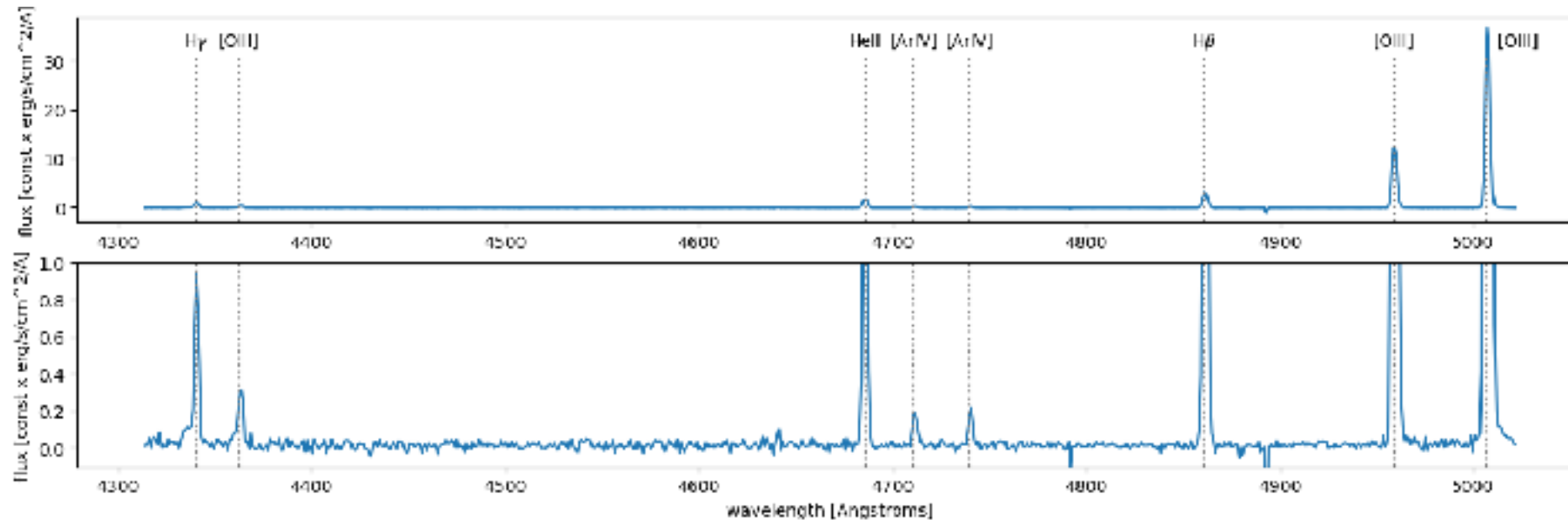
Lab 2

- Option A) Exoplanet transit
 - photometry of a star that hosts an exoplanet
 - extensive lab manual
 - new camera!
- imaging data analysis is easier (conceptually) than spectroscopic data analysis
- involves more programming later on (we're not doing photometry by hand 🤪)
- for Lab 3, have to then use spectroscopy



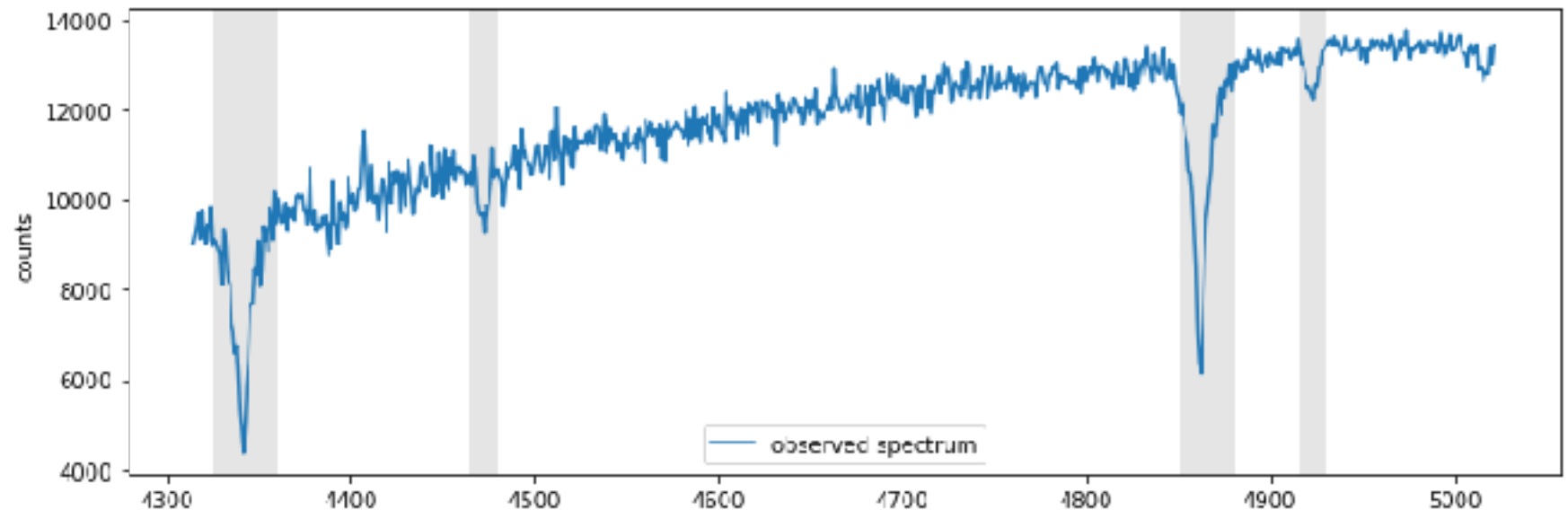
Option B.1) Spectroscopy of an emission-line nebula

- measure density and temperature of a planetary nebula



Option B.2) Spectroscopy of bright stars

- measure strength of absorption line features as function of temperature



Lab 2

- spectroscopic data analysis conceptually more involved than imaging data analysis
- somewhat less programming intensive
- for Lab 3, have to then do imaging

Lab 2

- Start discussing which Lab 2 option you want to choose
- Should have completed at least the day-time observations of Lab 1 before you decide
- Will look to scheduling these between now and October 9th.
- Once you have decided, let us know. THEN: select targets according to instructions and request 3 observing nights - first come, first serve